# Conformal campaigns: planning a round on counts, not rates

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## Abstract

Conformal prediction returns prediction sets whose coverage is guaranteed, but in a screening campaign that guarantee is a request rather than a finding: the level is chosen and the threshold is built to deliver it, so every setting is equally guaranteed and coverage ranks none of them. It cannot say which setting to screen under, how many compounds will be assayed, or how large a round must be. We keep the two class-wise requests as coordinates instead and state what each one does as counts of the outcomes a campaign is planned on: confident hits, false leads, missed in-class compounds, compounds assayed. Expectations of those counts cannot be planned on or constrained with a stated probability, so we bound them in finite samples from a labelled audit set drawn from the library to be screened, and invert the bound to read off the round size an audited count implies. The audit is a second step of the same form, exempting the calibration set as the first exempts the training set, so its counts survive a calibration that does not match the screen; what it requires instead is the chemistry the batch will contain. In a DNA-encoded library screened for soluble epoxide hydrolase, an audit matching only the batch's proportion of new chemistry covered $0.63 \pm 0.10$ of the consequence lines, no better than the $0.63 \pm 0.13$ of ignoring the shift altogether and no nearer the request; the two differ by $0.000 \pm 0.070$ on a paired comparison. An audit drawn from the batch's own building blocks covered $0.97 \pm 0.04$, but that arrangement splits one pool at random and so serves as a positive control rather than as evidence that a pilot repairs a broken campaign. Withholding blocks, the split used to guard against memorisation, cost 0.07 in ranking and two thirds of the coverage, so the ranking a screen reports does not announce the loss. Whether the match holds is testable beforehand from structures alone: the two preparations that fail read 0.66 against a permutation null centred on 0.50, and the one that holds reads 0.50. Dividing a pooled electrolyte compilation by the paper each measurement came from, rather than at random, drops coverage from $0.97 \pm 0.04$ to $0.70 \pm 0.12$, so provenance is part of what must be exchangeable. A batch the model chose is the hardest case and the one a self-driving laboratory commits: an audit reserved from the library could not even be formed in nine of twelve rounds, because nothing in-class was left to sample, while a random slice of the committed batch covered every line in all nine rounds where it could be. One round of 1500 candidates from a pool of 70 000 also took the in-class rate of the remainder from 0.0071 to 0.0013, so an audit prices the round it was drawn from and not the one after. Buying a pilot of the screened source repairs the provenance case: audited off-source, coverage is $0.55 \pm 0.07$; audited from a pilot of that source divided by concentration, by solvent system, or as the paper reported it — divisions that are not random and so can fail — it is 0.86 to 0.92. Quoted spreads are two standard errors over repeats, the repeat being the unit.

## 1 Introduction

A screening campaign commits a round at a time. A model scores a candidate library, a rule decides what goes into the next plate or run, and the round returns measurements that pay for the round after it. Increasingly the deciding is done by an automated or self-driving laboratory,[1,2] in which a model and a planner, rather than a person, choose what is made and measured. What such a platform is constrained by is integer and particular: the wells on a plate, the reagent budget of a run, the number of confirmed hits this round must return to justify the next. Those are statements about counts in one batch, not about rates in a population, and a campaign cannot be planned without them.

Distribution-free uncertainty quantification does not currently supply them. Conformal prediction wraps any trained model, needs only a labelled calibration set, and returns for each new input a prediction set containing the true label with probability at least $1 - \alpha$ whenever calibration and future data are exchangeable, with no assumption about the model or the distribution.[3,4] That generality has made it standard in machine learning and increasingly common in chemistry and materials science.[5–7] But the guarantee is coverage, and coverage is a level the team chooses and the threshold is built to deliver, so every setting is equally guaranteed and coverage ranks none of them. Risk-controlling methods move closer by bounding a chosen loss, but what they bound is a rate for one future candidate[8–10] (Section 1). Neither says how many wells this plate will consume or how many hits it will return.

The screens we consider decide which candidates go forward to a toxicity assay, a solubility or conductivity measurement, or synthesis. Because characterization runs in parallel, the capacity of a round is fixed before it starts. Unlike turn-by-turn optimization, a batch campaign often aims at covering a chemical or physical design space, mapping it or narrowing in on a set of optima (Section 4). The decision, whether made by a team or by a planner, is the same, and so is the gap we address. In a binary screen each compound receives one of four prediction sets: {out-class}, {in-class}, both labels, or neither. A team acts on these sets. It advances or discards compounds that receive a single label and sends the others for an assay.

To choose among these actions, a team needs to know what they will do: how many compounds will receive a single label, how many of those calls will be right, and how many compounds will be sent for testing. These consequences,

counted in compounds, are the terms in which screening decisions are made and budgets and capacities are planned. Coverage, the native coordinate of conformal prediction, is a statement about the prediction sets; it informs a decision once it is linked to these consequences. That link has to be measured. A set containing both labels is always correct but sends the compound to an assay, so two classifiers calibrated to the same coverage can sort a batch very differently.

Predictive uncertainty is ordinarily reported one prediction at a time, and reviews of uncertainty quantification for drug design and property prediction follow that convention. [11] Here the unit of prediction is the round rather than the compound, which changes what the uncertainty is made of. Two terms contribute to it: how well the audit pins down the rates, and how much the realized count varies around those rates in a batch of the size committed. The second does not shrink however large the audit grows, and at the batch sizes a plate imposes it is the larger of the two — which is why a guarantee stated as a rate does not answer the question, and why the construction below is stated in counts.

**What we do and why.** We give coverage a specific role. The coverage level is a setting the team chooses, separately for the in-class and out-class compounds, and changing it changes which compounds are advanced, discarded or assayed. We therefore treat the coverage request as the control: sweeping it moves the classifier through the space of possible decisions. A labelled *audit set*, a random sample of the library to be screened that was used neither for training nor for calibration, shows the consequences of each setting. From it we predict every screening outcome of a future batch, stated as counts of compounds, with prediction intervals that are valid in finite samples under exchangeability of the audit with the batch. That is a different requirement from the one coverage assumes, and Section 2.5 separates them: either can hold without the other.

These intervals are the deliverable, and obtaining them is what the method is for. Before a batch is screened, the audit states how many confident hits, false leads, missed in-class compounds and measurements the batch will contain, each as a count, equivalently a rate per 1000, with an interval that holds with a specified probability. The intervals are realistic in both directions: they hold at the stated level, where the usual shortcuts are too narrow, and they are no wider than the size of the audit requires. A team can therefore plan on them: book the assays, budget the measurements and expect the hits.

Coverage is thus the coordinate for navigating decisions, and the audited consequences are what a team chooses by. Choosing uses an objective, such as the expected cost or yield of the screen. It usually also uses guarantees: a plan the team can rely on, in which, with a stated probability, no batch needs more assays than the laboratory or the robot can run and every batch delivers enough hits to proceed. Such guarantees are bounds on counts that hold with a stated probability. The coverage requests reduce the navigation of these many consequences to two coordinates, in which a team can explore scenarios, each a set of such guarantees, and see which settings satisfy them.

Section 2 sets out the outcome table, the threshold correction, the navigation by coverage and the audit intervals that are the method's output; Section 3 validates the intervals in simulation and on real data and applies them to solubility, toxicity and electrolyte screens; Section 4 gives a workflow and the limitations.

**Contributions.** The statistical tools are classical. What is new is the separation they make visible, and how they are combined to answer the questions a screening team asks.

1. *A campaign is constrained in counts, and the guarantees in reach are not predictive.* The coverage level is chosen and the threshold is constructed to deliver it, so every setting in the grid is equally guaranteed and the guarantee ranks none of them. Risk-controlling methods bound a loss rather than deliver a request, but the object they bound is a rate for a single future candidate under a loss monotone in a scalar parameter, and a plate is constrained in integers: how many wells it consumes, how many hits it returns. At the batch sizes a platform commits, most of the uncertainty in those integers comes from the draw of the batch, which a bound on a rate does not describe. What the requests do supply is a coordinate system. Every screening quantity is a sum of cells of a $2 \times 4$ table of prediction set by true label (Table 1), and with class probabilities the class-wise requests reach the two-threshold decision rules on the model score, up to the spacing of the calibration scores and the resolution of the grid, so a sweep traces the decisions available to the team in two coordinates. Each setting induces a decision geometry of the same form as the cost-optimal reject rule where the probability of being in-class increases with the score, and the audited consequences show which costs it serves. That condition is not universal, and Section 3.5 reports the nine of twelve Tox21 endpoints where it fails and what that costs the reading.

2. *Two steps, two requirements, and only one of them is about the model.* Split conformal freezes a score fitted on training data and asks that calibration and future candidates be exchangeable. The audit repeats that move one level up: it prices a rule fixed by the training *and* calibration sets, and so exempts both. Coverage and predicted counts are therefore separate guarantees with separate requirements, and a campaign can hold either without the other. A rule calibrated on the wrong population, at the wrong prevalence, or against a cheaper label is a worse rule but not an invalid audit; an audit drawn from chemistry the batch does not share invalidates the counts however the rule was calibrated. We demonstrate both directions, and show that neither is visible in the metrics a screen reports: not in the ranking, which is prevalence-blind at these rates, and not in the enrichment of the confident calls, which does not order the designs by how badly they fail: withholding building blocks cost 0.07 in ranking and two thirds of the coverage.

3. *Predicted consequences with specified certainty.* For each setting, finite-sample valid prediction intervals and one-sided bounds for the future counts of confident hits, missed in-class compounds, false leads and deferred compounds follow from the audit set. [12,13] Expectations of these counts cannot be planned on and cannot be constrained with a stated probability, which is what a bound

supplies. The distinction is operational rather than technical: a plate has a fixed number of wells and a round must return enough hits to justify the next, so what a planner needs bounded is the count in the batch in front of it and not the rate of the population behind it. At plate scale most of the uncertainty in that count comes from the draw of the batch, which a guarantee on a rate does not describe. With a Bonferroni correction over the sweep the bounds remain valid for whichever setting is chosen, so a setting can be chosen under chance constraints on assay capacity, budget or yield. We verify them in simulation and on three real datasets: toxicity (Tox21), aqueous solubility (AqSolDB) and electrolyte conductivity (CALiSol-23).

4. *Sizing a round.* Inverting the same intervals answers the question asked before a campaign starts: how many candidates a batch needs for a target number of hits. The interval is valid for each batch size fixed in advance and the inversion makes the size a function of the audited count, so the figures are the size that count implies rather than a size carrying its own guarantee; Section 3.3 states this precisely. This builds on established estimates of campaign hit counts from calibrated probabilities [14] and extends them from an expectation to a bound. The distinction matters most where screens operate, since at low in-class rates the number of candidates required exceeds the expectation-based figure several-fold. The size of the audit enters that requirement directly, so audit measurements and synthesis trade against one another, and the trade speaks to the cost objection that led the expectation-based approach to forgo exchangeability in the first place. The two measurements enter the width differently: the audit fixes how well the rate is known, the batch fixes how much the realized count varies around it at a known rate, and the second does not shrink however large the audit grows. They are comparable when audit and batch are of similar size, so much larger audits buy little.

5. *Matching the audit to the screen, and checking the match.* Building the audit from the chemistry the round contains is therefore the design problem the guarantees reduce to. Where a round reaches into chemistry the audit does not hold, reproducing the batch's composition does not help. A pilot of the screened population does repair it, and we test that against divisions that are not random: a pilot divided by concentration band, by solvent system or as a paper reported its own measurements recovers coverage from 0.55 to between 0.86 and 0.92, against a ceiling of 0.99 for a random division of the same population. For a batch the planner chose, the same rule applies and is exact: a random slice of the committed batch is exchangeable with its remainder however the batch was selected, and covers where an audit reserved from the library cannot be formed at all. How large such a pilot must be for the count intervals to hold we do not establish: the attainability rule we give sizes a pilot used as a *calibration* set, and a pilot of new chemistry is drawn by whole blocks or series, so its in-class count is overdispersed relative to that rule. The requirement binds the calibration, audit and batch sets only, so a project may still separate its training compounds by scaffold or building block, and should. Whether the match has been achieved can be tested before the batch is made, from structures and without measurements, because two samples from one distribution cannot be separated better than chance by a classifier held out of its own training data. The test is the classifier two-sample test, [15] read against a null obtained by permuting the two sets; in cheminformatics the question it answers is the applicability-domain question, [16] answered by separability rather than by similarity. When scores are class probabilities, the calibration set can stand in for the audit set, at a quantified cost in width.

**Related work.** Conformal prediction is established in cheminformatics, for toxicity, bioactivity, imbalanced data, applicability domain and drift. [5–7,17–23] The closest work estimates what a screening round will return. Buendia *et al.* predict the number of hits an iteration of an iterative high-throughput screen will retrieve, by summing per-compound calibrated probabilities from Venn–ABERS predictors over the selected compounds, and show across six targets that the sum tracks the realized count; [14,24] Svensson *et al.* link conformal predictions to the gain of a screen. [25] Estimating campaign-level counts is therefore established, and we build on it. A sum of calibrated probabilities is an expectation, and we add a probability statement about the realized count, which is what a plan that has to hold requires. Buendia *et al.* also identify the obstacle to such a statement: selecting the highest-ranked compounds leaves the calibration and screened sets progressively less exchangeable, retaining validity would require measuring a fresh random subset of the screened library at every iteration, and the cost of those measurements would make the approach impractical. They therefore keep the initial calibration set, and report that the realized hit count falls outside their expected limits in later iterations. That randomly chosen, measured subset is the audit set used here, and its cost can be set against the compounds it saves (Section 3.3). These studies report coverage and efficiency, or a gain, on test data; we add prediction intervals for what a future batch will contain. In statistics, the coverage of a conformal predictor given its calibration set is known exactly, [26–28] and related to tolerance regions and PAC prediction sets. [29,30] Risk-controlling methods are the closest prior art and the comparison is worth making precisely, because they answer a different question rather than a weaker version of the same one. Conformal risk control bounds an expectation, $\mathbb{E}[L_{n+1}(\hat{\lambda})] \leq \alpha$, marginally over the calibration set and the test point together, for a single future point, under a loss required to be monotone in a scalar parameter. [9] The loss itself is general — it may be any bounded function, including an indicator that a plate has overrun its capacity, so the family is not confined to per-candidate error rates — but what is controlled is its expectation or its level, not the integer the plate returns. Risk-controlling prediction sets make that statement high-probability and conditional on the calibration set, $\Pr(R(\hat{\lambda}) \leq \alpha) \geq 1-\delta$; [8] learn-then-test extends it to several parameters and several risk constraints. [10] Song *et al.* come closest to the construction here, bounding the false discovery proportion of a conformal selection with a bound that holds simultaneously over all thresholds, so that a threshold may be chosen after the data are seen; [31] their envelope is

a sharper instrument than the Bonferroni correction we use for the same purpose, and the quantity it bounds is a proportion among the selections of the batch in hand rather than a count in a batch not yet measured. Bounding a realized count in a finite selected set is older than any of this and is not ours: the post hoc, true-discovery literature has given simultaneous high-probability bounds on the number of true discoveries in an arbitrary selected set since 2011.[32–34] What distinguishes the construction here is not that it counts but that it is *predictive*: those bounds describe the set in hand, from $p$-values computed on it, whereas the intervals below describe a batch not yet made or measured, from a sample of the library it will be drawn from. They are also two-sided, so a capacity and a yield constraint can be imposed at once; they cover every cell of the outcome table, including the missed in-class compounds, which a bound computed without labels cannot reach because a miss is defined by a label it never sees; and they invert for the batch size. The price is labels: an audit is measured, and the post hoc bounds are not.

A campaign is constrained in counts, not rates, and the two differ by more than a rounding. A guarantee that the long-run failure rate is at most 0.1 does not say that a plate of 96 will hold at most 13 failures with probability 0.95, because the realized count varies around the rate by an amount the rate bound does not describe. That variability is not a correction at the scale these platforms run: at an audit of 400 and a batch of 50, 89% of the variance of the interval reported here is contributed by the draw of the batch rather than by uncertainty in the rate (Section 3.3). A bound on the rate alone therefore leaves the larger part of that variability undescribed, and leaves most of it undescribed exactly where autonomous platforms operate, in small batches run often. A rate bound can of course be composed with a binomial tail to yield a count bound; the composition is valid and pays twice, once for the rate and once for the tail, where the construction below conditions on the total and pays once. The construction here bounds the realized integer count in a batch of stated size, two-sided so that capacity and yield can both be constrained, for every cell of the outcome table rather than one monotone summary, without any monotonicity requirement, and invertible for the batch size itself. Conformal selection[35] selects candidates with false-discovery control, and set-valued classifiers with bounded class-wise error[36] minimize ambiguity. Decision-theoretic work links prediction sets to actions. Conformal decision theory calibrates a decision parameter to a target risk,[37] and Kiyani *et al.* showed that prediction sets are a sufficient statistic for risk-averse decision makers with a known utility, for whom acting on the worst case within the set is optimal.[38] The question for which costs a classifier, or an abstaining classifier, is optimal goes back to the reject option[39] and to ROC analysis.[40,41] Our aim is complementary. We assume no utility, report *all* screening outcomes of each coverage setting with valid intervals and bounds, and let a team choose afterwards under its own objective and constraints.

# 2 Methods

## 2.1 Setting and assumptions

We call the class with the property the screen looks for *in-class* (label 1) and the other *out-class* (label 0): compounds active at an endpoint in Tox21, soluble compounds in AqSolDB, and electrolytes that reach the conductivity target in CALiSol-23. A trained binary classifier is held fixed. It gives each compound a predicted probability of being in-class, $u = \hat{p}(\text{in-class} \mid x)$. A labelled *calibration set* sets the thresholds. A labelled *audit set*, which in machine-learning terms is a held-out test set, is used only to measure what the thresholds do; we call it an audit set throughout, because what it is for here is pricing a round rather than scoring a model. A *future batch* of $m$ compounds will be screened. We assume throughout that the audit and future compounds are exchangeable with one another given the trained model and the calibration set, *i.e.* that their joint distribution is unchanged when they are reordered. The requirement is conditional on the calibration set, which need not itself be exchangeable with either; a random division of a fixed library satisfies it even though the three parts are then dependent through the library's finite size. Where the coverage guarantee is also wanted, the calibration set must in addition be exchangeable with the batch; the two requirements are separated in Section 2.5 and they fail independently. What must be exchangeable is each candidate together with the label recorded for it, not the candidate alone, so the assumption can fail in two ways that look nothing alike: the compounds may be drawn from different populations, or the compounds may be interchangeable while the label recorded on one set is not the label recorded on another. The first is a question of which compounds a round contains and the second of what is measured on them, and both appear below. Exchangeability holds when all three sets are random samples of the library to be screened and carry the same label. The training set is deliberately absent from that list. The classifier is held fixed and enters only as a score, so how its training compounds were chosen bears on how good that score is and not on whether the intervals are valid. The two splits a campaign makes are therefore independent and answer different questions. Separating the training compounds from the rest by scaffold, series or building block guards against a model that has memorised its training chemistry, and lowers an optimistic estimate of how well it ranks compounds unlike the ones it was fitted on. Dividing what remains into calibration, audit and batch at random is what keeps the intervals valid. Both can be done at once, and should be: the arrangement that measures a model on unseen chemistry *and* prices the batch correctly is a split at the training boundary followed by random assignment within the deployment population. The unit of sampling is the candidate to be screened: a compound, or for electrolytes a formulation, not an individual measurement. It fails when the batch comes from other chemical series or scaffolds (Section 3.12), and again when the thresholds are set against a cheaper label than the one the campaign's claim concerns (Section 3.11). It fails too when the batch is chosen using the model, as in active learning, unless that selection is part of the design. Section 3.9 tests that case and the two repairs available for it. The probabilities need not be calibrated in the usual sense; only their ordering matters.

## 2.2 Prediction sets and the outcome table

We use class-conditional (Mondrian) split conformal prediction,[3] which controls coverage separately for the in-class and out-class compounds. The label *out-class* is included in a

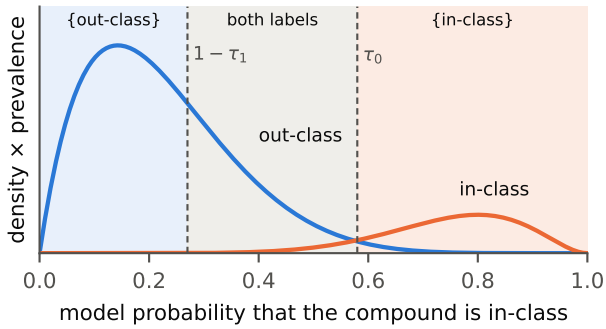



**Figure 1:** Class-conditional thresholds divide compounds into prediction sets. Curves show the distribution of the predicted probability of being in-class for out-class (blue) and in-class (orange) compounds, weighted by prevalence. Compounds left of $1-\tau_1$ receive {out-class}, those right of $\tau_0$ receive {in-class}, and those in between receive both labels. If the thresholds cross, the middle band receives the empty set instead. The coverage request for each class moves one threshold.

**Table 1:** Outcomes of a binary conformal screen. Each compound falls into one cell; every screening quantity is a sum of cells.

| True label | {out-class} | {in-class} | both | none |
|---|---|---|---|---|
| out-class | correct rejection | false lead | deferred | deferred |
| in-class | missed in-class | confident hit | deferred | deferred |

compound's prediction set when $u \le \tau_0$, where $\tau_0$ is set from the out-class calibration compounds. The label *in-class* is included when $u \ge 1-\tau_1$, where $\tau_1$ is set from the in-class calibration compounds (Fig. 1; ESI Section S1). Each compound then falls into one cell of a $2 \times 4$ table of true label by prediction set (Table 1). Every screening quantity is a sum of cells:

- missed in-class compounds are in-class compounds labelled {out-class};
- false leads are out-class compounds labelled {in-class}, *i.e.* compounds followed up that turn out out-class;
- deferred compounds receive both labels or none;
- coverage is the sum of the cells in which the true label is in the set;
- the hit rate of the confident calls is a ratio of two cells.

The cells in each row add up to the number of compounds of that class. Moving a threshold therefore moves compounds between cells of the same row, so the outcomes trade off against each other. With class probabilities, a compound can receive both labels or none, but not both kinds for the same thresholds (ESI Section S6). We report the two kinds separately, because an empty set signals a compound unlike the calibration data rather than an ambiguous one.[5]

## 2.3 Coverage requests and thresholds

The standard threshold guarantees coverage of at least $1-\alpha$ only on average over calibration sets. With tens of in-class calibration compounds, the coverage of the one model actually deployed can fall well below $1-\alpha$. The coverage observed over a batch of $m_y$ future compounds of class $y$ follows a Beta–Binomial distribution that depends only on the number of calibration compounds of that class and on the chosen order statistic. This result requires only exchangeability. It is the classical distribution-free rank law behind Wilks–Wald tolerance limits, read in the conformal setting;[26,28,29] the correction rule built from it is a two-parameter tolerance limit transposed to a finite batch. We choose the least conservative threshold for which batch coverage is at least $1-\alpha_y$ with probability at least $1-\delta_y$, separately for each class. A coverage request therefore has four numbers, $(\alpha_0, \delta_0, \alpha_1, \delta_1)$. The guarantee is stated for a planned number $m_y$ of future compounds of each class, which we set from the batch size and the class proportions of the calibration set. That is a plug-in for a quantity the campaign does not know, and at low prevalence it is estimated from few compounds, so the tolerance $\delta$ is controlled for the planned count rather than the realized one and the deployed guarantee is approximate in it; the correction is conservative in the count, and the real-data checks below came out conservative, but we have not bounded the error. In an automated laboratory the batch is a plate or one run of the platform, so $m$ is set by its capacity. The batch-free version,[26] which follows a Beta law, controls coverage only for an infinitely large batch (Section 3.1). As baselines we use the standard threshold and a one-sided correction based on the Dvoretzky–Kiefer–Wolfowitz inequality.[42]

Tied scores, for example from identical fingerprints, keep the coverage guarantee conservative, but they can break the count intervals that take a threshold from the same compounds they then count, because tied compounds cross it together; the audit intervals are unaffected, since the thresholds are fixed before the audit is counted. We therefore break ties at random: every compound receives an independent uniform number, and compounds are ordered by score and then by that number. This needs no knowledge of future scores and makes all the laws exact (ESI Section S2).

## 2.4 Navigating decisions with the coverage request

**Coverage as a coordinate.** With class probabilities, every rule that acts on single labels and defers the rest is a two-threshold rule on $u$: label out-class below one threshold, in-class above another, defer in between. The two coverage requests set the two thresholds, so a sweep of $(\alpha_0, \alpha_1)$ over a grid, at fixed $\delta$, traces the available two-threshold rules, up to the resolution of the calibration scores and of the grid. We keep one representative for each distinct pair of thresholds. Each setting maps to a vector of consequences, the counts of the outcome table and their bounds; the sweep traces a two-dimensional surface in this many-dimensional space, and its Pareto front over the outcomes that matter is the map on which a team explores its options.

**The decision geometry of a setting.** Without stated costs, a setting together with the usual rule (act on single labels, defer the rest) induces a decision geometry: two thresholds on the model score. Given only the prediction sets, this rule is the risk-averse (max–min) optimal use of the sets for any costs in which a deferral is cheaper than either error (ESI Section S7).[38] Its geometry matches Chow's cost-optimal reject rule[39] when the probability of being in-class increases with $u$, so each setting is expected-cost optimal for some

region of cost ratios, possibly empty. Which costs a setting serves depends on the fractions of in-class compounds within each prediction set, which labelled data reveal; for a known utility, sets built for that utility can serve it better.[38] The coverage request therefore locates a decision, and its audited consequences evaluate it.

**Objectives and chance constraints.** A team chooses a setting by an objective, such as the expected number of false leads, the expected cost or the expected yield, subject to constraints on what the campaign may consume or must deliver: at most $U$ assays, at most $U'$ missed in-class compounds, at least $L$ confirmed in-class compounds in a batch of $m$. Each constraint must hold with probability at least $1-\varepsilon$, so it is checked against a one-sided bound from the audit set, not against the expected count. This is a chance-constrained program[43] with two decision variables, solved by enumerating the grid, in which the constraint probabilities are replaced by distribution-free bounds from the audit. A *scenario* is such a set of constraints. Its feasible settings form a region of the coverage plane, and the objective selects a setting within it. When the objective and every constraint are monotone in the outcomes (fewer measurements, false leads or misses, and more confirmed compounds, are never worse), an optimal setting always lies on the Pareto front of the audited outcomes, taken over the quantities the objective and constraints use: a dominated setting is never better than the setting that dominates it and meets no constraint that setting fails. For every such objective, the table of Pareto settings with their expected counts and bounds is therefore a sufficient summary, and no cost ratio is needed to decide. Only an objective that is not monotone, such as a target workload that should be neither exceeded nor undershot, needs the full map. Changing the scenario, for example tightening the assay budget, moves or shrinks the region and shows what the change costs. An empty region is itself a result: the requirements cannot be guaranteed with the labelled data in hand, and either a constraint must be relaxed or more compounds labelled.

## 2.5 Prediction intervals for the consequences

**What the method returns.** For a chosen setting and a planned batch of $m$ candidates, the output is one line per outcome of Table 1: the expected count, and an interval that contains the realized count with a stated probability. Counts for a batch of 1000 are rates per 1000, so the same statement can be read either way. Tables 4 and 5 show the form. Everything else in this paper is built from these intervals: the sweep compares settings by them, the scenarios constrain them, and the audit-size rule plans their width.

**From the audit set.** Suppose an outcome (one cell, or a sum of cells) occurs $K$ times among $n$ audit compounds. Given the calibration set, the audit compounds and the $m$ future compounds are exchangeable. Given the total count, the number in the audit set is therefore hypergeometric, whatever the data distribution. Inverting this exact hypergeometric test gives conservative limits $[L,U]$ for the future count $K_m$ that hold with probability at least $1-\gamma$; they are quantiles of two Beta–Binomial distributions[12] (ESI Section S3). We use $\gamma = 0.05$ throughout. The construction holds for any outcome, any score and any fixed decision rule; it does not depend on conformal prediction. In the language of quality control the audit set is an acceptance sample, and the limits are the classical tolerance limits for the number of defectives in a future lot; the same logic, of bounding a finite population's count from a measured sample and inverting for how large that sample must be, is what makes a risk-limiting election audit work.[44,45] [29,46]

**Two steps, two requirements.** It is worth naming what that conditioning does, because it accounts for the results below. Split conformal fits a score on training data, freezes it, and asks that calibration and future candidates be exchangeable; the training set is exempt precisely because it enters only through the frozen score. The audit repeats the move one level up. The rule it evaluates — the score together with the thresholds — is fixed by the training and calibration sets, and is then frozen, so the audit asks that audit and future candidates be exchangeable and exempts both of the sets that built the rule. From the audit's point of view the calibration set is training data. The argument is the same one twice, though not the same formula: a rank is uniform and a split of a count is hypergeometric for the one reason that under exchangeability every ordering, and so every subset of a given size, is equally likely.

Each step therefore carries its own guarantee and its own requirement, and a campaign can meet either without the other. Coverage is the first step's guarantee and needs the calibration set to be exchangeable with the batch. The predicted counts are the second step's and need the audit set to be, whatever the calibration was built from. So a rule calibrated on the wrong population, at the wrong prevalence or against a cheaper label is a worse rule but not an invalid audit, while an audit drawn from chemistry the batch does not share invalidates the counts however well the rule was calibrated; both appear below. The same reading explains why the calibration set cannot freely audit itself: for the second step it is training data, and a rule may not be priced on the data that set it (Section 3.12).

**What the intervals cover.** The intervals cover the realized count in the future batch; no fixed underlying rate is assumed. They are conservative because counts are discrete. Each interval is valid for one outcome and one setting fixed before the audit set is seen. Choosing a setting by looking at its audited outcomes and then quoting its intervals is optimistic. Intervals computed at level $\gamma/J$ for each of the $J$ requests in the grid hold simultaneously, and therefore remain valid for whichever setting is chosen. For a scenario with $C$ bounded outcomes, each one-sided bound is computed at level $\varepsilon/(JC)$, so that every bound of every setting holds together with probability at least $1-\varepsilon$ and the chosen setting satisfies all constraints with at least that probability. Because the bounds hold jointly for all settings and all bounded outcomes, a team may pose any scenario over these outcomes, and change it, after looking at the map, without losing the guarantee. All intervals and bounds are conditional on the calibration set, that is, on the thresholds actually deployed.

**What makes an interval useful.** Validity alone is attained by the trivial interval $[0,m]$, which contains every possible count of an outcome in a batch of $m$ and says nothing. An

interval therefore earns its place by its width, and we report how many times narrower it is than that baseline, $m/(U-L)$, alongside coverage, following the principle of sharpness subject to calibration.[47] Width is set by the audit size and by how well the model ranks, and it can be planned before any labelling (Section 4). Whether the rule is better than no model at all is a separate question about the rule, not about a future count, and we test it on the audit set.

**From the calibration set alone.** With class probabilities and class-conditional thresholds, the calibration set can partly serve as its own audit. Whether an out-class compound receives the label *in-class* depends only on the threshold set from the in-class calibration compounds, so for this event the out-class calibration compounds are exchangeable with future out-class compounds and the audit construction applies to them. Whether an out-class compound *keeps* the label out-class is a coverage event, whose batch count follows the exact rank law used for the thresholds.[28] Each cell of a row is one of these two events or what remains after them, so computing both at level $1-\gamma/2$ bounds every cell of the row; a mixed batch adds an interval for the number of in-class compounds (ESI Section S4). Evaluating a class's own threshold on the compounds that set it, as is done when ordinary rates are computed on the calibration set, is not allowed; it makes the counts too narrow (ESI Section S5).

## 2.6 Experiments

**Simulations.** Eight designs are described in ESI Section S8:

1. audit-set intervals, with $10^4$ replicates, each drawing its own data distribution;
2. finite libraries: 20 fixed libraries, each split at random 1000 times, which gives exchangeable data that are not independent;
3. threshold correction, with absolute Cauchy scores, $n =$ 50–500, $m = 100$ and $10^6$ draws;
4. expected cost of the swept settings, computed exactly on the population, and selection: 4000 replicates of choosing among 100 settings under an assay capacity, with and without a Bonferroni correction;
5. scaffold dependence: compounds in clusters that share a shift of the predicted probability, with the future batch drawn from the calibration clusters or from new clusters;
6. tied scores: a cluster of identical scores at a threshold, with ties kept or broken at random;
7. calibration-only intervals: 3000 replicates per calibration size, compared with spending the same labelling budget half on calibration and half on an audit set.
8. a two-level draw, in which a rate is drawn and then the calibration, audit and future counts are drawn given it, used to separate a statement about the long-run rate from a statement about the realized count of one batch.

**Data.** Molecules were standardized with RDKit[48], which keeps the largest fragment, neutralizes it and writes canonical SMILES; duplicates were then merged. One model was used throughout: Morgan fingerprints[49] (radius 2, 2048 bits) and CatBoost gradient-boosted trees[50] (500 trees, depth 6, learning rate 0.1) with fixed hyperparameters. The experiments concern the coverage settings and their audited consequences for a given model, not the choice of model. For the twelve Tox21 endpoints,[51,52] each endpoint was split 50/25/25 into training, calibration and audit sets: 50 times at random and 20 times by Bemis–Murcko scaffold.[53] Under the scaffold splits we also drew pilot samples of 200, 500 and 1000 compounds at random from the audit scaffolds, used each as the calibration set, and screened the remaining compounds. For AqSolDB,[54] compounds with $\log S \geq -4$ were labelled soluble. Ten stratified 70/15/15 splits were made, the model was trained on all training compounds, and calibration and audit were restricted to $\log P > 3.5$. Solubility most often limits lipophilic compounds, so a screen of this subpopulation is a realistic use; it is also a subpopulation on which the model's scores are distributed differently from the training data. For CALiSol-23,[55] a compilation of measured conductivities of lithium-salt electrolytes, each formulation (salt, concentration and solvent fractions) was labelled good if its conductivity at 20–30 °C was at least 8 mS cm$^{-1}$. The same model was trained on all temperatures of the training formulations, with temperature, solvent fractions and fraction-weighted solvent descriptors as inputs, and scored each formulation at 25 °C. Four designs, 50 repeats each, drew calibration and audit sets from random formulations, from held-out salt–solvent systems, or, for a new salt, from the other salts or from a pilot sample of the new salt. To check the audit intervals on real data, each audit set was split at random, 20 times, into an audit half and a stand-in future batch. A random halving makes the two halves exchangeable by construction, so these checks verify that the construction and its implementation are right and cannot fail for any model, representation or label noise; they are not evidence that an audit prices a batch assembled some other way. The designs that test that are the block hold-outs and the declared round, where the batch is a distinct set of molecules. For the DNA-encoded library,[56] a sample of one million molecules was taken from the released table and a molecule was labelled in-class if it carried the library's own binding call for soluble epoxide hydrolase, a rate of 0.74%; the analysis runs on a further random subsample of 200 000 of those molecules, at a rate of 0.72%. Each molecule is built from one building block at each of three positions, so designs were defined over blocks rather than molecules: assignment at random, and assignment of whole blocks at one of the three positions, so that a batch molecule carries a block the training molecules do not. A declared round, 30% of it built on first-position blocks absent from training, was then prepared for in three ways that differ only in where the calibration and audit sets draw their novel chemistry. Counts for this library are for the realized batch of each repeat, tens of thousands of candidates, because at a sub-percent rate a batch of 1000 carries too few in-class molecules for a confirming setting to be bounded away from zero; the scripts additionally scale each rate to a nominal $10^5$ so repeats can be compared, and those scaled figures are not the ones quoted here. The thresholds for this library were planned at $m = 1000$ rather than at the realized batch size, which makes the deployed rule more con-

servative than the request rather than less; ESI Section S14 gives the consequence, and it should be repaired before the library results are relied on. A further division of the electrolyte formulations by the paper each measurement came from, rather than at random, is described in ESI Section S10. Details, counts and seeds are in ESI Section S9, and for the library in ESI Section S14.

# 3 Results

## 3.1 Predicted counts hold at the stated certainty

The audit predicts the count of each outcome in a future batch, with an interval meant to hold with probability 95%. The checks in this section impose exchangeability by construction, through random splitting, and so establish that the construction is correct rather than that it survives a real change of population; the sections that put that to the test come later. In simulation, the realized count in a batch of 1000 compounds fell inside its interval in 95.7–98.9% of replicates, for every outcome and every audit-set size (Fig. 2). The intervals were also tight: for wrong single-label calls (false leads plus missed in-class compounds) they spanned about $\pm 51$ per 1000 with 100 audit compounds and $\pm 20$ with 1000 (ESI Table S6). The same held when fixed libraries were split at random, a design that is exchangeable but not independent. Averaged over libraries, coverage was 96.1–98.9%, and no library fell below 94.7% (Monte Carlo standard error 0.7% per library; ESI Table S5). The usual plug-in interval, Binomial quantiles at the audited rate, treats that rate as if it were known; it covered only 44–75% of batches with 100 audit compounds and 84.5–93% with 1000, and fell to 23% for the worst library (ESI Tables S6 and S5). The opposite shortcut, $m$ times the Clopper–Pearson interval for the rate, ignores the sampling variation of the future batch: it held with 100 audit compounds (96.3–98.6%) but fell to 94.0–97.6% with 300 and 86.6–94.3% with 1000. Each shortcut omits one of the two sources of variation, the uncertainty of the audited rate and the variation of the batch, and fails where the omitted one dominates (ESI Section S8); the audit interval accounts for both. Scaled Clopper–Pearson intervals are therefore adequate only when the audit set is much smaller than the batch, and not for small batches such as a single plate. Bayesian posterior predictive intervals with a uniform prior were 1.3–5.9% narrower and close to nominal in these designs, but carry no guarantee; in a two-level design they fell to 94.8% (ESI Table S7). That is a trade rather than a refutation: a fifth of a point of coverage against a few percent of width, against an exact interval that is itself conservative by one to four points in the same tables. The audit interval held at its nominal level throughout. Width is what distinguishes these intervals from the trivial $[0, m]$: for a batch of 1000 the 95% intervals of the four constrained outcomes at the chosen setting of each scenario screen below spanned 16–109 counts, 9 to 63 times narrower than that baseline, and 36–204 after the joint correction over the sweep, 5 to 28 times narrower.

On real data the audit intervals behaved the same way (Table 2). Splitting each Tox21 audit set into an audit half and a stand-in future batch, the intervals covered 95.6–99.0% of the realized counts, for every outcome. Per endpoint, coverage was 95.8–97.1% under random splits. Those are pointwise figures, one outcome at a time, and the object a team is handed is the whole table: all eight lines of one setting — the six cells and the two aggregates — held together in 82.7% of random splits and 81.9% of scaffold splits. Eight pointwise 95% intervals do not make a 95% table, and a team that needs the table jointly must divide the level by the number of lines it reads. It was 95.0–97.5% under scaffold splits, in which the model had been calibrated on other scaffolds: the audit described the consequences correctly because it came from the compounds being screened. On AqSolDB, over all 132–144 settings of the sweep and all outcomes, the pointwise intervals covered 96.1% of 216,960 realized counts. With the Bonferroni correction over the 144 requests, every interval of every setting covered its count together in 98.5% of 200 audit–future splits.

**Table 2:** Real-data check of audit intervals (nominal 95%). Each Tox21 audit set (1405 to 1755 compounds) was split 20 times at random into an audit half and a stand-in future batch of the same size; entries are the percentage of realized future counts inside the interval. Random: 600 splits, 12,000 checks per outcome. Scaffold: 240 splits calibrated on other scaffolds, 4800 checks per outcome.

| Outcome | random | scaffold |
|---|---|---|
| correct rejection | 96.2 | 95.6 |
| false lead | 95.9 | 96.2 |
| deferred out-class | 95.7 | 95.6 |
| confident hit | 96.6 | 96.5 |
| missed in-class | 99.0 | 99.0 |
| deferred in-class | 96.9 | 96.4 |
| single label | 95.8 | 95.8 |
| deferred | 95.8 | 95.8 |

**Thresholds for small calibration sets.** With the standard threshold, coverage over a batch of 100 compounds fell below the requested 90% in 35–42% of calibration draws for $n = 50$ to 500, against a requested tolerance of $\delta = 10\%$; the batch correction brought this to 4.7–9.9%, matching theory to within Monte Carlo error (ESI Table S1). The batch-free correction is identical for $n \leq 100$ but too permissive beyond it, falling short in 11.8% of draws at $n = 150$ and 21.2% at $n = 500$, so the batch form should be used whenever batch performance matters; the one-sided DKW correction was valid but far more conservative. The same holds on real data: per-class audit coverage fell below 90% in 55 of 600 random Tox21 splits for out-class compounds and 45 for in-class compounds, against 56.2 and 42.3 expected from the exact law for the thresholds used (ESI Table S12), where the batch-free correction fell short in 17% and 13% of splits and the DKW correction in 0.2% and 1.0%.

## 3.2 What accepting the default setting costs

Table 3 compares, on the population, the expected cost of three choices: the best two-threshold rule without constraint, the best swept setting with coverage of at least 90% for both classes, and the default setting with coverage of exactly 90% for both classes. Thresholds are set at the exact quantiles, without a finite-sample correction, so the table isolates the effect of the coverage setting itself. The constrained optimum cost exactly as much as the unconstrained one in 46 of

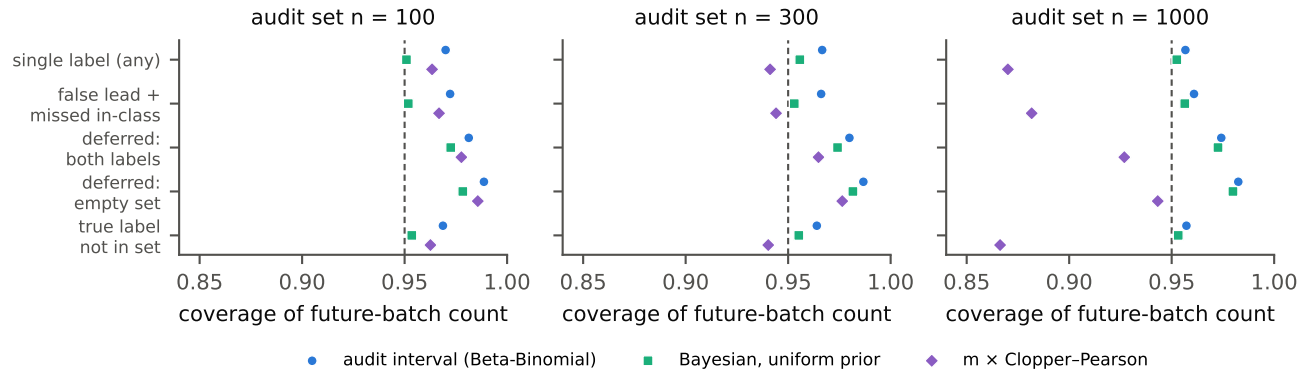



**Figure 2:** Coverage of 95% prediction intervals for the count of each outcome in a future batch of 1000 compounds, from an audit set of $n = 100$, 300 or 1000 compounds (panels); $10^4$ replicates, each drawing its own data distribution. Audit interval: the Beta–Binomial limits of Section 2.5. Bayesian: posterior predictive with a uniform prior. $m\times$Clopper–Pearson: the interval for the rate scaled to the batch, which ignores the sampling variation of the batch. The plug-in Binomial interval, which ignores the uncertainty of the audited rate, covered only 44–93% and is omitted (ESI Table S6). Dashed line: nominal 95%. Monte Carlo standard error is about 0.2%.

60 settings and within half a percent in a 47th, and at most 47% more; where it cost more, the unconstrained optimum misclassified more than 10% of one class. The default setting, chosen without the costs in view, cost more. With a miss ten times as costly as a false lead, it cost up to 3.1 times the optimum for $\rho \geq 0.3$. When an assay was nearly free ($\rho = 0.1$), it cost up to 7.3 times the optimum at $\lambda = 10$, 13.6 times at $\lambda = 20$ and 32 times at $\lambda = 50$. The sweep of coverage settings thus contained the optimal rule in most cases, and navigating it with the costs in view finds that rule. That is a statement about the sweep, not a claim to beat a rule built for a known cost: where the costs *are* known and the probabilities can be recalibrated, a cost-optimal reject rule is the thing to use.[39] On the solubility screen the swept setting is indistinguishable from such a rule built on isotonically recalibrated probabilities, costing between 0.009 less and 0.013 more per compound across four cost regimes against standard errors of 0.004 to 0.038, and it is dearer by 0.011 to 0.171 than one built on Platt scaling (ESI Table S13). In the most lopsided regime, a miss fifty times a false lead and an assay half of one, both are beaten by deferring every compound: 0.665 per compound for the swept setting and 0.674 for the isotonic rule against 0.500 for deferring everything, which is a property of that cost structure rather than of either rule but is worth stating plainly. The case for the sweep is therefore that it needs no cost ratio and returns audited counts for whatever is chosen, not that it is cheaper when a ratio is in hand.

Choosing the setting from audited outcomes needs the correction for selection. Among 100 swept settings, we chose the one with the fewest audited errors whose upper limit on deferrals fitted an assay capacity of 15% of the batch (ESI Section S8). With the Bonferroni correction over the settings, realized deferrals exceeded capacity in 0.03% of replicates; without it, in 0.47%. Both are below the nominal 5%, because neighbouring settings have strongly correlated counts, but only the corrected choice carries a guarantee. The correction cost 52.5 rather than 43.5 errors per 1000 compounds.

## 3.3 Sizing a round

A campaign asks the audit's question in reverse. A project needs a certain number of hits to continue, and what it must decide is how many candidates to put in the batch. The smallest batch whose lower bound on confident hits reaches the target answers that, and it follows from the same intervals by a search over the batch size. The interval is valid for each batch size fixed in advance, and the search makes the size a function of the audit count, so the probability statement does not transfer unchanged; because the lower bound increases with the batch size the selected size is the one at which the bound first reaches the target, which makes the reading natural but is not a proof, and we have not established a statement uniform in the batch size. The figures below should be read as the batch size the audited count implies, not as a size carrying its own guarantee.

**Table 3:** Expected cost relative to the best two-threshold rule (%), for three score distributions. $\lambda$: cost of a miss relative to a false lead; $\rho$: cost of an assay relative to a false lead. Constrained: best setting with population coverage $\geq 90\%$ for both classes. Default: population coverage exactly 90% for both classes. Exact population values, without the finite-sample correction. Full grid in ESI Table S3.

| | $\rho = 0.3$ | | $\rho = 0.8$ | |
|---|---|---|---|---|
| Scores, $\lambda$ | constr. | default | constr. | default |
| well separated, 5 | 0 | 12 | 0 | 58 |
| well separated, 20 | 0 | 3 | 0 | 9 |
| overlapping, 5 | 0 | 31 | 6 | 5 |
| overlapping, 20 | 0 | 100 | 0 | 16 |
| balanced, 5 | 0 | 91 | 12 | 35 |
| balanced, 20 | 0 | 453 | 9 | 203 |

The answer differs from the one an expectation gives, and the difference is largest where screens operate (Fig. 3, left). Dividing the target by the observed in-class rate, which is what a sum of calibrated probabilities amounts to,[14] asks for 1000 candidates to obtain 20 hits at a rate of 2%. The batch whose audited count implies 20 hits at a confidence of 0.95 holds 2056 candidates when the rate was measured on an audit of 384, and 1687 on an audit of 1000. At a rate of 50% the expectation asks for 40 and both audits give 52. The gap closes as the rate rises, because what the bound carries is the uncertainty in the rate itself, and that matters least when the rate is high and the audit has seen many in-class compounds.

The size of the audit enters the requirement directly, which prices audit measurements against synthesis. At a rate of 2%, an audit of 49 compounds leaves the required batch at 19 091 candidates and an audit of 384 at 2056, so 335 further audit measurements remove about 17 000 syntheses from the requirement; at a rate of 10% the same 335 measurements remove 201. Those are the requirements at the in-class count each audit is expected to see, and at these rates that count is small enough that the requirement is better read as a distribution. An audit of 49 at a rate of 2% sees one in-class compound with probability 0.37, giving 19 091, and none with probability 0.37, in which case no batch size reaches the target at all; two gives 2783. At an audit of 384 the count has a mode of 7 or 8 and the requirement runs from about 22 000 at two in-class compounds to 900 at fifteen, with a median of 2056, which is the figure quoted at the expected count, and a mean of about 3500 that the long upper tail carries. The comparison between audit sizes is therefore a comparison of distributions: enlarging the audit from 49 to 384 removes the chance of an unanswerable round and roughly halves the typical requirement, which is the claim the figure supports, rather than a fixed saving of 17 000 syntheses. On this accounting the audit is not an overhead on the campaign but the measurement with the largest effect on what the campaign costs, and its effect is largest exactly where a screen's hit rate is low.

Three quantities are stacked in that figure, and separating them is what makes it planable. Were the rate known exactly, the batch implied for 20 hits at a confidence of 0.95 would still hold 1390 candidates rather than 1000, because which candidates fall in this particular batch is not something any amount of measurement can settle. Knowing the rate only from an audit of 384 raises the requirement to 2056, and knowing it from an audit of 49 raises it to 19 091. The first gap is the risk the campaign carries, the second and third are what ignorance of the rate costs on top of it, and only the latter two are for sale. A team that plans on the expectation omits all three.

How much of that is worth buying can be settled before any compound is measured, because the width of an interval has a closed form in the audit and batch sizes (ESI Section S3). Accepting a width $r$ times the narrowest any audit could achieve requires $n = m/(r^2-1)$ audit candidates, a ratio that does not depend on the rate or on the batch size: an audit the size of the batch lands a factor $\sqrt{2}$ above that floor, one of $2.3m$ within 20% of it and one of $4.8m$ within 10%. The floor itself does depend on the rate, and at the rates a primary screen works at it is wide — for a batch of 1000 at a rate of 2% it is ±9 on an expected 20 — so a team should read the floor first and enlarge the batch, not the audit, when the floor is what binds. Correlated series change these two statements differently. The floor is a width, so where candidates fall in correlated series it is inflated by the square root of the design effect; enlarging a batch with analogues of series it already contains adds little to its effective size.[57,58] The ratio rule $n = m/(r^2-1)$ needs no inflation at all, because the design effect multiplies both variance terms and cancels from $r$. A rule that sizes a *sample* rather than a width, such as the attainability rule of Section 3.12, inflates by the design effect itself.

The exchange has a limit (Fig. 3, right). An interval's width carries one term in the audit size and one in the batch size (Section 2.5): the first is the uncertainty in the rate, which the audit reduces, and the second is which candidates happen to fall in this particular batch, which it cannot. The two contribute equally when audit and batch are of the same size, and beyond that the batch term dominates, so an audit much larger than the batch buys little. For a batch of 50, an audit of 400 already leaves 89% of the interval's variance attributable to the batch.

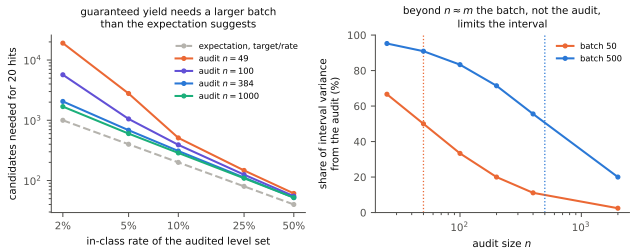



**Figure 3:** Sizing a round. Left: candidates the audited count implies for 20 hits at a confidence of 0.95, against the in-class rate of the audited level set, for four audit sizes; the dashed line divides the target by the rate, as an expectation does. Right: the share of an interval's variance contributed by the audit rather than by the batch, with the dotted lines marking where the audit matches the batch in size.

The limits used here are one-sided: the sizing search inverts a lower limit placing the whole of $\gamma = 0.05$ in the lower tail, not the two-sided limits of ESI Section S3, which place $\gamma/2$ in each. The factor between the two conventions depends on the audit: it is about 1.1 at an audit of 1000 and 1.2 at 384, and reaches two only at the smallest audit considered here, where 19 091 becomes 38 693. The convention is therefore part of every figure quoted.

Re-estimating between rounds costs little, with two conditions. A campaign that screens a first batch, revises the rate and sizes the next has looked at the data more than once, which a fixed-sample interval does not cover; dividing the confidence level across a declared number of looks restores a simultaneous statement over those looks. Three looks rather than one raise the required batch from 309 to 350 candidates at a rate of 10%, and from 2056 to 2605 at 2%, so declaring three looks in advance costs between a seventh and a quarter of a batch, 27% at the lower rate. The conditions are that the number of looks is fixed before the campaign starts, and that the correction is over the looks only. Stopping when the hits so far look sufficient is not covered: the continuation is then data-dependent, and an interval reported at the stopping look is conditioned on having reached it. A campaign that wants to stop when it likes needs an anytime-valid construction — an alpha-spending boundary or a confidence sequence — rather than a division of the level,[59–61] which we do not develop here. Nor does any of this repair the selection in the inversion itself, discussed above; it addresses only the multiplicity of looking more than once. Which data the revised rate comes from also matters, and if it comes from a batch the model selected, the revision inherits that selection.

## 3.4 What a measurement budget costs: a solubility screen

After standardization and de-duplication, 9254 AqSolDB compounds remained, 2467 of them with $\log P > 3.5$. In

each of ten splits, the lipophilic calibration and audit sets held about 370 compounds each, 22% of them soluble ($\log S \geq -4$). Audit AUROC was 0.91–0.93 overall and 0.82–0.88 within the lipophilic subset.

**What the labels can carry.** The solubility labels are a cut through a continuous property measured across sources, so they are noisy where it matters, at the cut. Taking the spread between sources as its scale, about 8% of the screened compounds near the cut would be labelled the other way by another laboratory, which is the size of the requests being swept; a sweep read at that resolution has to be shown to survive it. We therefore re-ran the whole analysis on the same model and the same scores, changing only the labels: moving the cut half a log unit each way, flipping 8% of the compounds within half a log unit of it at random, flipping the same number in order of their recorded disagreement between sources, and flipping those the score ranks most extremely against their class.

Two of the three claims survive and one does not. The resolution of the sweep survives: the number of distinguished settings stays between 132 and 144 under every perturbation but the upward shift of the cut, which lowers it to a median 120. The enrichment survives in sign and in significance, at a fold between 1.51 and 3.61, significant after the Bonferroni correction in every one of the hundred perturbed evaluations — so an enrichment of about twice the base rate is safe, though the two digits of 2.4 are not. What does not survive is the scenario, which is the part a team would plan on. The number of certifiable settings falls from a median 26 to 18 under random flips and to 2 and 0 under the two adversarial variants, with six of ten splits admitting no certifiable setting at all once the most discordant labels are flipped, and the cheapest certifiable budget rises by a factor of 2.2 to 2.8. The Pareto front keeps its size but not its membership, retaining a median 74 to 85% of the settings it held under the recorded labels and as little as 44% in the worst split. A quarter of the front is therefore an artefact of the labels, and a scenario chosen at this audit size and this label quality should be read as a demonstration of the machinery rather than as a plan.

One detail of this is worth stating because it cuts against the intuition the proxy-label section below establishes. Flipping the compounds the score ranks most extremely *flatters* the screen — more settings become certifiable, the enrichment rises, the budget falls — because noise correlated with the score makes the labels agree with the model. The damaging variant is its mirror image, flipping the compounds the score ranks least extremely against their class, and that is the one reported above. Score-structured label noise is therefore not uniformly ruinous: which way it is structured decides whether it hurts, and a proxy that agrees with the model is the one that hides its own failure.

**The price of a budget.** Cutting the measurement budget from 450 to 250 per 1000 lipophilic compounds cost 73 additional false leads, with all requirements guaranteed jointly at 95%, and the cheapest budget any setting can be guaranteed to meet is 173 measurements per 1000, at which exactly one setting remains certifiable and its guaranteed false leads rise to 335. At this audit size the guarantee itself cost about 110 measurements per 1000: the chosen setting expects 286 and is bounded at 395. Those figures are one split, and the spread across the ten is wide enough to matter: the number of certifiable settings runs from 11 to 65 and the cheapest certifiable budget from 29 to 309 per 1000, with a median of 134. The setting chosen here is itself certifiable in only four of the ten splits, so it is this split's choice and not a recommendation. The rest of this section shows how these numbers follow from the map.

**The decision space.** Sweeping $(\alpha_0, \alpha_1)$ over 12 values each, with $\delta = 0.10$, gave 132–144 distinct settings per split. Fig. 4 (left) shows their audited consequences for one split: one model, one endpoint, and outcomes ranging from 10 missed soluble compounds per 1000 with a third of the batch sent to measurement, to almost no measurements at the price of about 50 missed soluble compounds. Of the settings, 74–120 per split were Pareto-optimal for missed soluble compounds, false leads and measurements (ESI Table S10). A front that large invites the question whether it is a decision surface or a restatement of the grid, and the answer depends on the objective: minimising missed compounds selected the same request in eight of the ten splits and one two grid steps away in the other two, while minimising deferrals wandered over eight distinct requests, all within one corner of the grid. The front is therefore stable where the objective is steep and loose where it is flat, and a team should read the chosen setting's audited counts rather than its coordinates. So the front is the menu, and the team's constraints make the choice. Table 4 gives the audited consequences of the two extremes, with intervals for a batch of 1000. Setting A misses 10 soluble compounds per 1000 and sends 335 to measurement; setting B sends none to measurement and misses 52. Because A and B were picked by looking at the audit, the intervals that hold for them are the Bonferroni ones of Table 4, $[0, 49]$ and $[15, 114]$ missed, rather than the pointwise $[2, 29]$ and $[29, 84]$ that the same table reports for a setting fixed in advance. Over the ten splits the corresponding audited values were 0–20 missed and 203–522 deferred for setting A, and 26–61 missed and 0–42 deferred for setting B.

**Scenarios.** Two scenarios illustrate the choice (Fig. 4, centre and right). In both, a batch of 1000 lipophilic compounds must yield at least 60 confirmed soluble compounds and miss at most 90, and the objective is the fewest expected false leads. In the first, at most 450 compounds may be sent to measurement; in the second, at most 250. The limits were set after inspecting the map, which the joint bounds allow. Every constraint was checked against a one-sided audit bound, with the Bonferroni correction over the 144 requests and the four bounded outcomes, so that all constraints hold together with probability at least 95%. Under the first budget, 26 of the 144 requests were feasible; the chosen setting expects 115 false leads, 286 measurements and 23 missed soluble compounds per 1000, with bounds, holding together with probability at least 95%, of 395 measurements, 75 misses and at least 66 confirmed. Tightening the budget to 250 left 9 feasible requests and moved the choice to a setting expecting 188 false leads: the tighter budget costs 73 additional false leads per 1000. Choosing instead by the upper bound on false leads, a risk-averse objective, selected the same settings in both scenarios. Both objectives are monotone, and as expected every chosen setting lay on the Pareto front of the four bounded outcomes, which held 105 of the 144 settings (ESI

Section S7); the Pareto table is a complete menu for such objectives, and the constraints select from it. Over a range of budgets the price of the budget can be read off directly: the best feasible setting expected 221 false leads at 200 measurements per 1000, 188 at 250–300 and 133 at 350, and beyond 375 the budget no longer constrained the choice (115 false leads). Four settings remained certifiable at 200 and one at 173, below which no setting met all requirements with the labels in hand, so a team wanting a tighter budget must relax a requirement or label more compounds. The joint correction is not free in reach as well as width: over the ten splits the share of the grid that satisfies the scenario's limits falls from 0.2–0.6 when each bound is read pointwise to 0.0–0.1 when they must hold together, so a scenario that looks comfortably feasible on point estimates can have almost no certified settings. Against the trivial interval $[0, 1000]$, which validity alone attains, the chosen setting's 95% intervals were 9 to 26 times narrower for the four constrained outcomes, and 5 to 14 times narrower after the joint correction over the sweep. The advanced compounds, those given a single in-class label with no measurement spent, were 54% soluble against a prevalence of 22%, an enrichment of 2.4 times, on 51 of 95 compounds that is significant on the audit set after a Bonferroni correction over the 144 settings (one-sided Fisher exact test, $p = 2 \times 10^{-15}$), so the model demonstrably earned the measurements it saved. The gap between expected and bounded counts (286 against 395 measurements) is the price of the guarantee at this audit size, and labelling buys it down: at the same audited rate, half the audit (192 compounds) would give a bound of 432, twice the audit 372 and four times 358. A capacity rule chosen this way on one half of the audit set and checked on the other half never exceeded capacity in 200 splits (median upper bound 484, realized 304 measurements per 1000). No cost ratio is needed for this choice; the cost ratios for which each setting is expected-cost optimal are mapped in ESI Fig. S2.

## 3.5 One request, twelve sets of consequences: Tox21

After standardization and de-duplication, 7592 Tox21 compounds remained, with 5619–7017 labelled per endpoint and an in-class prevalence of 2.9–15.9% among labelled compounds (the labels are missing non-randomly, which the Limitations quantify). Here a confident hit is a compound confidently flagged as toxic, which a screen would set aside. Calibration sets contained a median of 44–228 in-class compounds per endpoint (ESI Table S12). Audit AUROC was 0.70–0.87 for random splits and 0.67–0.85 for scaffold splits. These endpoints also bound the decision-geometry reading of Section 2.4, which needs the probability of being in-class to increase with the score. It does not here: the in-class fraction decreases over at least one adjacent pair of score bins in nine of the twelve endpoints, nominally so for two at $z = 2.4$ and 2.6 — an uncorrected comparison that does not survive correction across the 108 bin pairs examined — and for three endpoints the fraction is nearly flat over the lower nine bins with the signal confined to the highest (ESI Table S9). The containment argument and the cost-optimal reading therefore do not apply to those endpoints, though the intervals themselves need no such condition. Before tie-breaking, a median of 3.3–4.3% of scores were tied, mostly from identical fingerprints.

**The same request, very different consequences.** All endpoints were calibrated with the same request, $\alpha = \delta = 0.10$ per class, using the batch correction. Yet the median fraction of compounds receiving a single label ranged from 13.6% (NR-PPAR-$\gamma$) to 62% (NR-AhR) (Fig. 5). Table 5 gives the audited consequences of the two extremes, for the split with the median single-label fraction. For NR-AhR, a batch of 1000 compounds yields 78 confident hits (audit interval 58–101) against 78 false leads (57–100). For NR-PPAR-$\gamma$, it yields 15 (6–26) against 89 (67–113), with 865 compounds deferred. Half of the toxic calls are correct for NR-AhR, one in seven for NR-PPAR-$\gamma$. The audit reveals these consequences before the screen is run. The fraction of in-class compounds among single-label "in-class" calls ranged from 13% to 58%, an enrichment of 3.0 to 6.5 times over prevalence. With the standard threshold the single-label fractions were larger, 28% and 72%, but that threshold's coverage of a class fell below the 90% target in 41.5% of the 600 endpoint–split pairs for in-class compounds and 47% for out-class compounds, and in 54–80% of splits per endpoint for at least one of the two.

## 3.6 Auditing a new salt: an electrolyte screen

CALiSol-23 holds 13,825 conductivity measurements of lithium-salt electrolytes from 27 publications.[55] Three source counts appear below and each belongs to a different pool: 27 in the released file, 26 after the exclusions described next, and 25 contributing formulations measured at room temperature, which is the pool every experiment here uses. After removing 523 rows without a conductivity and 23 rows from one source whose values lie about 100 times below all others for the same formulations (a probable unit error), 1495 formulations had been measured at room temperature, 275 of them (18%) at or above $8\,\mathrm{mS\,cm^{-1}}$. The threshold is close to the conductivity of standard carbonate electrolytes; the ten room-temperature measurements of 1 M $LiPF_6$ in EC/EMC (3:7) in this dataset run from 7.85 to $10.21\,\mathrm{mS\,cm^{-1}}$, so the reference electrolyte itself straddles the cut and a good formulation is one within reach of that standard rather than reliably above it. That spread is the price of a binary label on a continuous property measured across sources, and a quarter of the formulations (374 of 1495) sit within 20% of the cut. A threshold of $10\,\mathrm{mS\,cm^{-1}}$ would leave 102 good formulations in the pool, about 24 in a calibration set of 350, which is just above the attainability floor (Section 3.12) and would leave the request barely reachable. Measuring conductivity is cheap compared with building and cycling cells, so here a deferral means a measurement.

**A new salt.** The sharpest test holds out a whole salt. It is a weaker test than the name suggests: the salt held out is $LiPF_6$, which is the incumbent of commercial practice, supplies 606 of the 1495 room-temperature formulations, and anchors the class boundary, so what is tested is transfer to a well-represented salt rather than to novel chemistry; its share of good formulations, 25%, also differs from the salts calibrated on, so prevalence moves with the chemistry here as it does in the library screen. A genuinely novel salt is not available in this compilation, where several candidates contribute

**Table 4:** Audited consequences of two settings of the lipophilic solubility screen (split 0), per 1000 compounds: realized rate per 1000 in the audit set (384 compounds; 10 per 1000 is 4 observed compounds), with the 95% prediction interval for a batch of 1000 and the interval with Bonferroni correction over the 144 requests of the sweep, which remains valid for one nominated outcome after choosing a setting by its audited outcomes. Holding all six lines of a chosen setting jointly would require the level divided by $144 \times 6$ instead; the scenario bounds quoted in the text are computed that way, over the four outcomes they constrain. A: the Pareto setting with fewest audited missed soluble compounds, ties broken by fewer deferrals, $(\alpha_0, \alpha_1) = (0.30, 0.075)$; B: the Pareto setting with fewest audited deferrals, ties broken by fewer misses, $(0.275, 0.25)$; $\delta = 0.10$.

| | Setting A | | | Setting B | | |
|---|---|---|---|---|---|---|
| | audit | 95% | Bonferroni | audit | 95% | Bonferroni |
| missed soluble | 10 | [2, 29] | [0, 49] | 52 | [29, 84] | [15, 114] |
| false lead | 232 | [183, 285] | [148, 330] | 221 | [174, 274] | [139, 319] |
| soluble confirmed | 177 | [134, 226] | [103, 269] | 172 | [129, 220] | [99, 263] |
| insoluble rejected | 245 | [195, 299] | [159, 345] | 555 | [495, 614] | [446, 660] |
| deferred, insoluble | 299 | [246, 356] | [206, 404] | 0 | [0, 11] | [0, 26] |
| deferred, soluble | 36 | [17, 64] | [7, 92] | 0 | [0, 11] | [0, 26] |

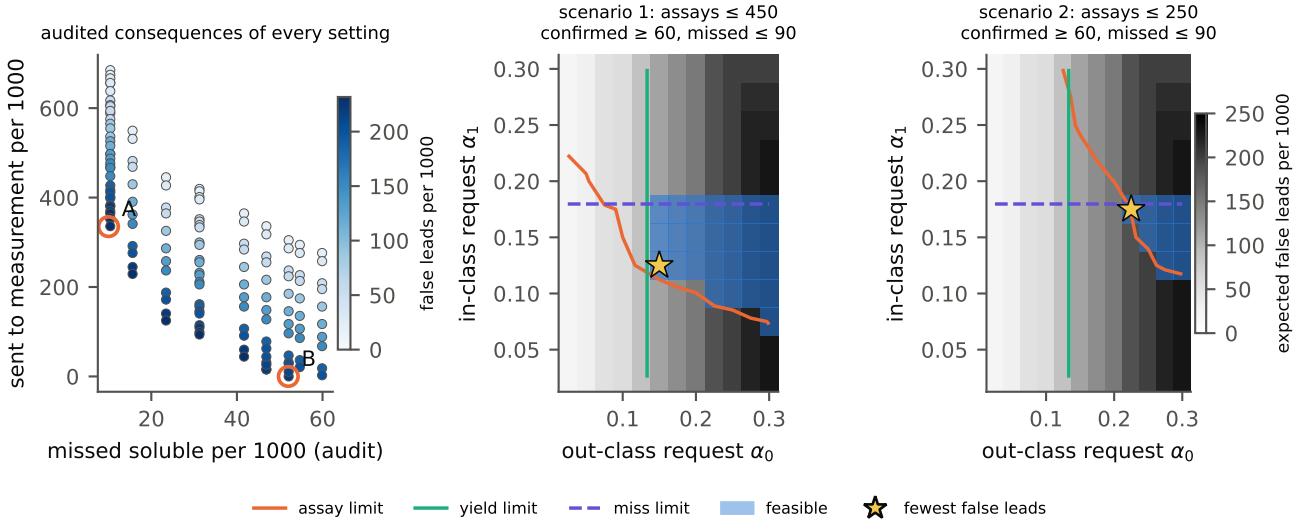



**Figure 4:** Navigating a solubility screen for lipophilic compounds (AqSolDB, $\log P > 3.5$; split 0) with one model. Left: each point is one setting of the coverage requests, placed by its audited missed soluble compounds and measurements, with false leads in colour; A and B are the settings of Table 4. Centre and right: two scenarios in the plane of the coverage requests, with the expected number of false leads (grey, the objective) for each request. Lines mark where the 95% joint bound of each constrained outcome equals its limit; blue cells satisfy all constraints with joint probability at least 95%, and the star is the feasible setting with the fewest expected false leads. The right panel tightens the assay budget from 450 to 250 per 1000.

a single room-temperature formulation each. With a model trained and calibrated on the other salts and the 606 $LiPF_6$ formulations screened, coverage of the good class averaged 29% and fell below 90% in all 50 splits (Table 6; Fig. 6, right). The audit of the screened formulations showed what this meant for the screen: at the default setting it reported a median of 174 missed good electrolytes per 1000 formulations, about two-thirds of the good ones, and 59 confident hits. Calibrating instead on a random pilot of 40% of the $LiPF_6$ formulations, about 240 measurements, restored it for the good class (5 shortfalls against 3.3 expected) and left a residual excess for the poor class (8 against 4.0, one-sided Poisson-binomial $p = 0.04$), and the audit showed the gain: 10 missed good electrolytes and 159 confident hits per 1000, for 308 measurements. Formulations split at random over all salts met the request (2 and 4 shortfalls against 4.7 and 3.1 expected), and the sweep of coverage requests gave 108–132 distinct settings per split, trading measurements (0–230 per 1000) against false leads (0–150) and missed good electrolytes (0–30) (Fig. 6, left). Holding out whole salt–solvent systems also broke the coverage guarantee (20 and 22 shortfalls against 4.5 and 5.0).

**The audit carries the guarantee.** In all four designs, including the two in which the coverage guarantee failed, an audit set from the screened formulations reported the consequences correctly. Audit intervals for the default setting covered 95.5–100% of realized counts, pointwise intervals over the whole sweep 97.2–97.6%, and with the Bonferroni correction every setting and outcome was covered together in 98.4–99.2% of audit–future splits. A capacity rule chosen on one audit half, as a planner would choose it, never exceeded capacity on the other half, in 980–1000 checks per design. The intervals were also informative: for the scenario setting the 95% intervals were 12 to 63 times narrower than the trivial $[0, 1000]$, and 7 to 28 times narrower jointly over the sweep. The advanced formulations were 71% good against the 15% prevalence of the audit set of the split these maps use, which is lower than the 18% of the room-temperature compilation as a whole, an enrichment of 4.7 times (one-

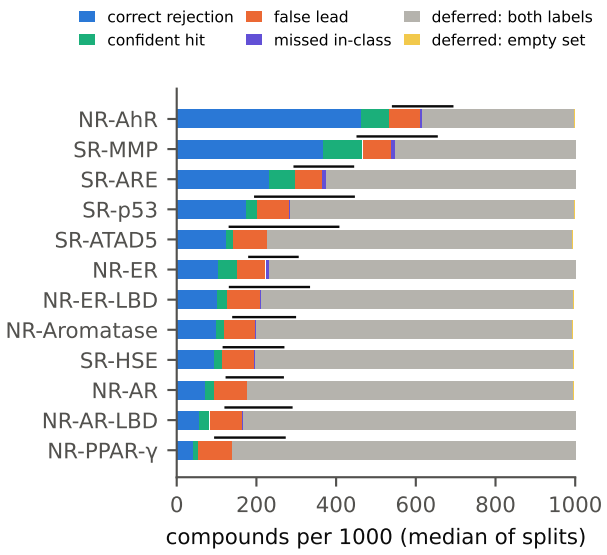



**Figure 5:** Twelve Tox21 endpoints calibrated to the same request ($\alpha = \delta = 0.10$ per class, batch correction) sort a screening batch very differently. Bars: median audited counts per 1000 compounds over 50 random splits, ordered by single-label fraction (medians need not sum to 1000); no empty sets occurred. Black lines: 5–95% range of the single-label count over splits.

**Table 5:** Audited consequences per 1000 compounds for the endpoints with the highest and lowest single-label fraction (batch correction), for the random split with the median single-label fraction (25th of 50). Entries: realized rate per 1000 in the audit set, with the 95% audit interval for a batch of 1000 in brackets. Calibration sets: 1385 out-class compounds and 197 in-class compounds (NR-AhR), 1526 and 37 (NR-PPAR-$\gamma$).

| True label | Outcome | NR-AhR | NR-PPAR-$\gamma$ |
|---|---|---|---|
| out-class | correct rejection | 456 [417, 496] | 32 [19, 48] |
| out-class | false lead | 78 [57, 100] | 89 [67, 113] |
| out-class | deferred | 342 [305, 380] | 850 [820, 877] |
| in-class | confident hit | 78 [58, 101] | 15 [6, 26] |
| in-class | missed in-class | 7 [2, 15] | 0 [0, 3] |
| in-class | deferred | 39 [24, 55] | 15 [6, 26] |

sided Fisher exact test on the audit set, $p = 2 \times 10^{-37}$, Bonferroni over 144 settings). Scenario maps for the electrolyte screen behave as for solubility (ESI Fig. S3). Intervals from the calibration set alone covered 99.7% of counts for random formulations; for a new salt they rely on a calibration that does not match, and a separate audit is the tool to use.

**Per plate.** A self-driving platform measures plates rather than batches of 1000. For the pilot-calibrated rule, an audit of 364 formulations gives, for one 96-well plate, 30 expected measurements with a 95% interval of 20–40; scaled Clopper–Pearson would give 25–35, too narrow at this ratio of audit to batch size (Section 3.1). The one-sided bound of 38 measurements per plate holds with probability 95% for a single plate, and a bound of 43 holds for all ten plates of a campaign together (level 0.5% per plate).

## 3.7 An audit and a batch that are not two halves of one set

Every check of the audit intervals reported so far divides one labelled pool at random, which makes the two halves exchangeable by construction and establishes that the construction is right rather than that it survives a real separation. The electrolyte compilation lets the harder question be asked, because it records which paper each measurement came from. We therefore divided the formulations by source instead of at random, so that no paper contributes to both the audit and the batch, holding the model and the calibration set fixed across the divisions and halving nothing. The protocol is the one used elsewhere in this paper: formulations are assigned to training, calibration and the remainder, the model is fitted on every temperature row of the training formulations and is therefore disjoint from the set that sets its thresholds, and the remainder is divided three ways.

The intervals do not survive it. Over ten repeats, random halving covered $0.97 \pm 0.04$ of the consequence lines, as it must; partitioning the papers at random and letting each formulation follow its own paper covered $0.70 \pm 0.12$; and taking one paper's formulations as the batch and the rest as the audit, over the five papers large enough to serve within the remainder, covered $0.77 \pm 0.11$. The inference here is about papers, so the paper is the unit and not the repeat: the five per-paper means run from 0.58 to 0.90, and pooling the 41 runs instead would quote $\pm 0.05$ and hide that spread. A division by provenance therefore costs roughly a quarter of the coverage a random halving delivers, and nothing a campaign can compute beforehand announces it. Area under the curve on the batch ran from 0.81 to 1.00 across these divisions. The enrichment of the confident calls, taken over the pool's base rate, which is the rate a campaign knows, sits between 3.3 and 4.9 in all three divisions and does not track coverage within any of them (Spearman $+0.01$ by source, $-0.19$ one source out). The one quantity that does track coverage is the ratio of the audit's in-class rate to the batch's, and that ratio needs the batch's labels, which is what the audit was to replace.

What the designs do not do is separate the causes. The median ratio of audit to batch in-class rate is 1.02 halved, 0.67 by source and 1.88 one source out, and the per-repeat ratios span 0.3 to 2.2 and 0.4 to 8.1 in the two source divisions. So dividing by provenance moves the in-class rate as well as the mapping from score to outcome, and the coverage loss is attributable to the two together. That is itself the practical point — a paper is a population in both senses — but it means this experiment shows that provenance is enough to void the intervals, not that the mapping alone is.

**A pilot of the right population, divided for real.** The remedy this suggests is the one the library screen also points to: when a round reaches into a population the audit does not hold, buy a pilot of that population. Every arrangement reported so far that recovers the nominal level divides one pool at random, which cannot fail, so none of them tests that advice. The compilation lets it be tested, because a single paper's formulations can be divided in ways that are not random. Holding the thresholds on the other sources throughout, so that the rule is calibrated off-population exactly as a campaign would have it, we held out one source as the screened

**Table 6:** Electrolyte screen (CALiSol-23, conductivity $\geq 8\,\mathrm{mS\,cm^{-1}}$ at room temperature), 50 splits per design. Shortfalls: splits in which audit-set coverage of the class fell below 90%, observed/expected under the exact law. Audit coverage: percentage of realized counts in the stand-in future batch covered by 95% audit intervals, for the default setting (range over outcomes), for every setting of the sweep (pointwise), and for all settings and outcomes together with the Bonferroni correction over 144 requests.

| | | Shortfalls | | Audit coverage (%) | | |
|---|---|---|---|---|---|---|
| Calibration and audit drawn from | AUROC | poor | good | default | sweep | Bonferroni |
| random formulations | 0.98 | 2/4.7 | 4/3.1 | 95.5–99.4 | 97.2 | 98.4 |
| new salt–solvent systems | 0.91 | 20/4.5 | 22/5.0 | 96.6–98.9 | 97.6 | 99.2 |
| new salt, calibrated on others | 0.92 | 0/4.5 | 50/1.9 | 95.9–100.0 | 97.3 | 98.9 |
| new salt, pilot calibration | 0.92 | 8/4.0 | 5/3.3 | 96.0–99.2 | 97.2 | 98.5 |

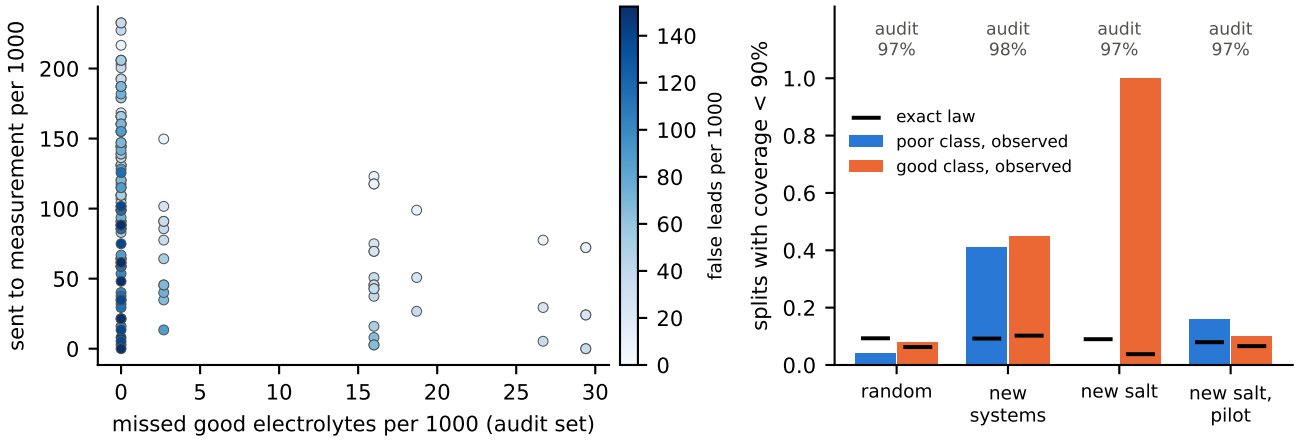



**Figure 6:** Electrolyte screen (CALiSol-23). Left: each point is one setting of the coverage requests for one random split, placed by its audited missed good electrolytes and measurements per 1000 formulations, with false leads in colour. Right: fraction of 50 splits in which coverage of each class fell below 90%, observed (bars) and expected under the exact law (black marks), for calibration on random formulations, on other salt–solvent systems, on other salts, and on a pilot sample of the new salt; above each design, the pointwise coverage of the audit intervals over the sweep.

population and varied only where the audit came from: from the other sources, as a team would do by habit; or from the held-out source itself, divided by concentration band, by solvent system so that no system is shared, or simply as the paper reported its formulations, first half against second.

The advice holds. Taking the source as the unit, an audit drawn from the other sources covered $0.55 \pm 0.07$ of the consequence lines over eight sources. A pilot of the screened source covered $0.92 \pm 0.17$ when the division was by concentration, $0.87 \pm 0.21$ by solvent system and $0.86 \pm 0.12$ as reported; on the sources that support both arrangements the paired gain over auditing off-source is $+0.37 \pm 0.21$, $+0.37 \pm 0.26$ and $+0.34 \pm 0.17$. A random division of the same source covered $0.99 \pm 0.02$, which is the ceiling the construction guarantees and which these divisions do not quite reach: sampling the right population recovers most of what provenance costs, and not all of it.

Two details are worth keeping. The divisions by concentration and by system leave the audit and batch in-class rates close, at median ratios of 1.04 and 1.09, so this is not a prevalence effect; the off-source arm sits at 0.56. And the division as reported, which is not chemically meaningful but merely not random, leaves the rates badly mismatched at a median ratio of 1.81 and still recovers to 0.86. Sampling the population the batch comes from therefore helps even where the prevalence does not match, which is the asymmetry the construction predicts: what the audit must share with the batch is the mapping from score to outcome, and a prevalence gap it can report.

Two things follow. The first is a limit on this work's own evidence: the coverages quoted for solubility, toxicity and electrolytes under random splitting are verification, and the only divisions in this paper that test the intervals against a genuinely distinct batch — these, and the library's block holdouts — break them. The second is a limit on pooled compilations, which is where much of the data for materials screening comes from. An audit drawn from the library to be screened is not sufficient if the batch will be measured by someone else, on another instrument, with another calibration: provenance is part of what must be exchangeable, and a compilation of 25 papers is 25 populations. The instruction to audit the population you will screen therefore extends to the laboratory that will screen it.

## 3.8 Building the audit for the round it prices: a DNA-encoded library

The electrolyte screen showed that an audit drawn from the screened population bounds the consequences where a calibration set from elsewhere does not. A campaign poses a harder version of the same question, because the next round is often not drawn from the population already screened: it introduces chemistry that has not been made. What the audit must then contain is a question about its design, and a DNA-encoded library answers it cleanly, because the chemistry is built from a catalogue of building blocks and a round can be

declared in those terms.

We use a sample of a published library[56] screened against soluble epoxide hydrolase, in which each molecule is assembled from one block at each of three positions and carries the library's own binding call; the in-class rate is 0.72% in the 200 000-molecule sample analysed here. That call is derived from selection counts, so it is itself the kind of proxy label Section 3.11 shows can be ruinous when its errors are structured with respect to a model's score; the coverage figures of this section are therefore statements about the recorded call and not about confirmed binding. At that rate a batch of 1000 contains about seven in-class molecules, too few for a setting confirming a handful of hits to be bounded away from zero, so counts in this section are given for the realized batch of each repeat, tens of thousands of candidates, rather than per 1000 as elsewhere.

How much is at stake can be measured in such a library by withholding blocks. Assigning molecules to training, calibration, audit and batch at random keeps all four exchangeable, and the intervals then covered 53 of 54 consequence lines over nine repeats, at their nominal level. Withholding instead every molecule carrying a given block at one position, so that the batch is built on chemistry absent from the calibration and audit sets, breaks them: over ten repeats each, coverage fell to 48% when the withheld position was the third, 37% when it was the second and 35% when it was the first. The six consequence lines of one repeat are cells of a single table on a single batch, so they are not independent checks and the repeat is the unit; taking it as such gives $0.48 \pm 0.19$, $0.37 \pm 0.14$ and $0.35 \pm 0.12$ against $0.98 \pm 0.04$ for random assignment, at ten repeats for each block design and nine for random assignment. The ranking metric does not announce this. The area under the receiver operating characteristic curve stayed between 0.71 and 0.98 throughout, but at a sub-percent in-class rate that statistic is governed by the ordering of the negatives and is invariant to prevalence by construction, so it cannot announce a prevalence change and its steadiness is not evidence that the failure is invisible. Nor does the enrichment of the confident calls rescue the diagnosis, though it is the statistic a screening group reports as a matter of course. Taken over the audit's own in-class rate it runs between 8.7 and 10.6 across the nine random-assignment repeats and between 1.4 and 17.5 across the ten first-position repeats, which looks like a signal until one notices two things: that enrichment so defined is the audit's in-class rate re-expressed, since it divides the precision of the confident calls by exactly that rate, and that within the first-position repeats it does not order them by coverage — its largest value, 17.5, belongs to a repeat covering three of six lines, while the repeat covering none sits unremarkably at 6.4. Significance is no help either: it is significant in every repeat of every design, including the repeat at an enrichment of 1.4, because the batch holds tens of thousands of candidates. The honest reading is that no statistic available before the batch is measured orders these designs by how badly their intervals fail. The audit set does announce part of it, and we did not use that signal. In-class molecules are concentrated in a few blocks — five first-position blocks carry 60.5% of the binders in the million-molecule draw, one of them at a rate of 0.709; in the 200 000 analysed the same five carry 61.1% and the richest binds at 0.685, against the sample's 0.0072 — so partitioning whole blocks moves the in-class rate of the audit and of the batch independently, between 0.2% and 2.5% across these repeats. A team holding a labelled audit at 1.6% in a library whose base rate is 0.72% is told by that number alone that the two sets do not match, before any model is consulted. How far that diagnostic goes is worth stating exactly, because it is weaker than the single repeat suggests. Across the thirty block-holdout repeats, the distance of the audit's rate from the library's predicts how many lines are covered only moderately (Spearman $-0.38$, $p = 0.04$), and within the ten position-1 repeats alone it predicts nothing ($-0.05$). What does predict coverage is the distance between the audit's rate and the *batch's* ($-0.68$ pooled, $-0.90$ at position 1), and that is the comparison a campaign cannot make before it screens. So the audit-against-library check is a cheap partial warning rather than a test, and it is the honest reason the failure should be called hard to see rather than invisible. The coverage failures reported here are therefore a change in prevalence as well as a change in the mapping from score to rate, and the designs do not separate the two: the arm that holds, drawing the audit from the batch's own blocks, matches the batch's prevalence by construction. Separating them would need block assignments stratified to equalize the two rates. Dealing the blocks to the four roles in order of their in-class count would keep the design block-disjoint while equalizing the rates, and is the experiment that would separate them; we have not run it. What the published deal gives, for reference, is an audit at 1.19% against a batch at 0.35% in its first repeat, which is the mismatch to be removed; and comparing the audits in-class rate against the library's is a diagnostic a campaign should add whatever that comparison shows. In the position-1 repeat whose promise overshot by the widest margin — not the worst by coverage, which is the repeat that covers none of its six lines — a batch of 42 145 candidates was promised between 578 and 723 confident hits, at a model AUROC of 0.89, and returned 119. That batch held 150 in-class molecules in all, so the interval was already refuted by the comparison of the two in-class rates, 1.62% in the audit against 0.36% in the batch, before the model was consulted; this is the signal of the preceding paragraph, and it is the reason the failure should not be called invisible. The same audit prices a nominal batch of 1000 at 8 to 24 confident hits, which is a separate interval computed at that size rather than this one rescaled.

The modest size of that drop is itself the point, because withholding blocks is the kind of split a project adopts deliberately. Separating the held-out compounds by scaffold, series or building block is standard practice for checking that a model has learned chemistry rather than memorised its training set, and read as such the result here is reassuring: mean ranking fell only from 0.960 under random assignment to 0.892, 0.936 and 0.935 as the first, second and third positions were withheld, while over the same runs interval coverage fell from 98% to 35, 37 and 48%. The ranking figures are means of dispersed quantities — the position-1 repeats run from 0.708 to 0.976 against 0.950 to 0.968 for random assignment — and at a sub-percent in-class rate the area under the curve is governed by the ordering of the negatives and is deliberately blind to the prevalence changes these designs introduce, so it understates how little the ranking reveals. The two move together, the largest ranking cost accompanying the worst coverage, but they move on scales that are not comparable: a drop of 0.07 is one a project would accept as evi-

dence that its model transfers, and it sits beside a guarantee that has lost two thirds of its coverage. Passing the check is therefore not evidence that the predicted counts transfer: it de-biases an estimate of how well the model orders compounds, and leaves untouched the mapping from score to rate, which is what a batch of new chemistry changes and what the intervals depend on.

A round is declared in which 30% of the batch is built on blocks at the first position that the model has never seen. Three ways of preparing for that round are compared, each supplying its own calibration and audit sets, all three sharing one model per repeat: calibration and audit drawn only from blocks the model trained on, as a team would do by habit; drawn with the same 30% share of new chemistry as the batch but from different new blocks, which is the position of a team that has measured some new chemistry and is about to make different new chemistry; and drawn with the same share from the batch's own new blocks, which is a pilot of the chemistry the round will contain.

Two features of the design bound what the comparison establishes. The naive and mirror arms are priced on one and the same batch, so their contrast is exact and paired. The pilot's batch is not that batch: it draws its novel 30% from all of the new blocks rather than from the half the other two use, so its batch is a different draw in every repeat and, where the pool runs short, between 1% and 9% smaller in four of the ten. More importantly, the pilot arm splits a fixed pool at the molecule level into calibration, audit and batch, which makes those three exchangeable by construction. Its 0.97 is therefore a positive control — the same verification a random halving supplies, and a result that could not have come out otherwise — rather than a demonstration that a pilot repairs a broken campaign. What the experiment establishes is the negative beside it.

Reproducing the batch's composition is worth nothing. Taking the repeat as the unit, as above, calibration and audit holding only familiar blocks covered $0.63 \pm 0.13$ of the consequence lines and the same sets carrying the batch's proportion of new chemistry from other blocks covered $0.63 \pm 0.10$. Because the two arms share a batch and nearly nested audits, their difference can be taken per repeat, and it is $0.000 \pm 0.070$ — a much tighter null than the two means compared separately would give, and the strongest form of the claim. Neither comes near the nominal 95%. Sampling the batch's own blocks covered $0.97 \pm 0.04$, and was the only arrangement to cover all six lines in a majority of repeats, eight of ten against one and none. The pilot's interval on confident hits was also the narrowest, spanning 70 candidates against 77 and 76, but that is not a property of the arrangement: the pilot's audit holds 24 349 compounds against 17 045 and 17 315, and 174 in-class against 120, and the width formula above accounts for essentially the whole difference (a predicted ratio of 0.911 against an observed 0.900). At a matched audit size the three widths would be indistinguishable. The valid arrangement therefore cost about 7300 extra audit measurements, which is the price the section's conclusion describes rather than an exception to it.

What distinguishes the three is therefore not how much new chemistry the audit contains but whether it contains *that* chemistry. The distinction matters because the first is cheap and can be arranged before synthesis, while the second requires measuring a sample of the new blocks, which is an experiment. Where a round introduces chemistry the audit does not hold, the two arrangements that do not sample the batch's own chemistry both fail, and no arrangement short of sampling it was found to work.

## 3.9 The batch the model chose

Every division so far has been random or structural. A campaign is neither: a planner reads the scores and commits the top of the ranking, which is the one arrangement this construction is not expected to price. We tested it on the library, holding the model fixed so that only the selection moves. A pool of 70 000 candidates was scored, the planner committed the highest-ranked $m$ of those remaining in each round, and the same committed batch was priced three ways: with a random sample of the candidates still in the pool, which is the audit a team would reserve; with a random slice of the committed batch itself, measured and used to price the remainder; and with the same selection rule applied to a pool of 20 000 reserved from the planner. Two round sizes were run, $m = 8000$ and $m = 1500$.

The audit reserved from the library does not price a plate the model chose, and the reason is visible before any interval is computed. In the first round at $m = 1500$ the committed batch had an in-class rate of 0.269 against 0.0071 for the pool it was drawn from, a factor of 38; what remained of the pool fell to 0.0013. Across the twelve rounds of the two campaigns the library audit could not even be formed in nine, because a sample of the leftover pool contained no in-class candidate at all, and in the three rounds where it could be formed it covered 2, 2 and 4 of the six consequence lines. A quantity that cannot be estimated is a safer failure than one estimated wrongly, but neither is a guarantee.

Both repairs work, and the first is exact. A random slice of the committed batch covered all six lines in every one of the nine rounds in which it could be formed, which is what the construction requires: a random division of a committed set is exchangeable with the rest of that set however the set was chosen, so the selection drops out. Applying the selection rule to a reserved pool also worked, covering all six lines in eight of ten rounds and four of six in the other two; it is an approximation rather than an identity, since the reserved pool and the screening pool are two draws rather than one, and it costs a pool of high-scoring candidates the planner may not commit. The three rounds in which the batch was rich enough for the comparison to carry weight — batch rates of 0.269, 0.055 and 0.020 — show the pattern most clearly: library audit unavailable, 2 of 6 and 2 of 6; batch slice 6, 6 and 6; reserved selection 6, 4 and 6.

The second finding is about the sequence rather than the round, and it is the one that bounds what any audit can do for a campaign. One round exhausts the library. Committing 1500 of 70 000 candidates, 2% of the pool, took the in-class rate of the remainder from 0.0071 to 0.0013, and committing 8000 took it to 0.0005; by the fifth round of the smaller campaign no arrangement could be formed at all, because nothing in-class was left to sample. So a round's audit does not price the round after it even with the model frozen, the rule fixed and the selection unchanged, and not because the mapping from score to rate has moved: the population underneath has been removed. An audit is a measurement of the library as it stands, and at a sub-percent in-class rate a single

round changes the library it measured. The instruction that follows is to reserve a slice of every committed round rather than to amortize one audit over a campaign, which is cheap, and to read the falling rate of the remaining pool as the campaign's own statement about how much is left to find.

## 3.10 Checking exchangeability before the batch is made

The preceding result is a design instruction, and a team following it would like to know whether it has succeeded. Exchangeability binds the calibration set, the audit set and the batch, and all three are sets of structures that exist before any of them is assayed, so the question can be put to the structures alone. We ask it as a two-sample problem: a classifier is trained to tell audit compounds from batch compounds using fingerprints and no labels. That is the pair the predicted counts depend on, since the audit prices whatever rule it is handed and asks only to be exchangeable with the batch; the calibration set may be compared with the batch in the same way when the coverage guarantee is also wanted. Under exchangeability the two are draws from one distribution, no classifier can separate them, and its area under the curve sits at one half.

Run across the four split designs, the test separated every arrangement that later failed and did not separate the one that held (Table 7); what it can refute is bounded by the representation, as the end of this section and the Limitations set out. It sits at 0.498 for random assignment, which covered 98% of consequence lines, and at 0.985, 0.989 and 0.997 for the three block holdouts, which covered 48, 37 and 35%; comparing the calibration set with the batch instead gives 0.501, 0.987, 0.990 and 0.999, so on these designs either pair would have raised the alarm. Beside it we report the share of batch molecules whose block at each position also occurs in the comparison set, which is the quantity the design controls directly. That share is one everywhere under random assignment and falls to zero at exactly the withheld position in each holdout, so the two statistics together give a verdict and name the axis responsible.

Those four designs are the easy case, and the table should be read as a check that the test fires rather than as a measure of its sensitivity: a withheld block catalogue is a substructure a fingerprint represents directly, so a classifier must find it, and the block-share column gives the same verdict by inspection. The question that matters is whether the test sees the three preparations of the declared round, which differ not in how much novel chemistry the batch carries — all three hold 70% of their batch's first-position blocks, or all of them for the pilot — but in where the audit draws its own. Over three repeats the audit-against-batch area under the curve was 0.657 for the naive preparation and 0.660 for the mirror, against 0.500 for the pilot; over the ten repeats of the coverage experiment the first two cover 38 of 60 consequence lines each and the pilot 58 of 60. The permutation null, obtained by shuffling the audit and batch labels and refitting, averaged 0.500 over 120 replicates and spanned 0.470 to 0.524, so the two preparations that fail are separated from the null and the one that holds is not. The test therefore distinguishes arrangements that a fingerprint comparison of composition cannot, which is what the workflow step requires of it.

The test is one-sided and should be read that way. Separation is strong evidence against exchangeability, because two samples from one distribution cannot be told apart. Failure to separate is weak evidence for it, since a classifier may simply lack the features to see a difference that matters; the fingerprints used here cannot distinguish stereoisomers, for instance, whose in-class rates in this library differ severalfold. A screen that passes is therefore not certified, but a screen that fails is refuted, which is enough to be worth running: it costs no measurements, uses structures already known, and can be run before the plate is committed. The clearest case it cannot see is the provenance division of Section 3.7: which paper a measurement came from is not a property of the formulation, so no test on structures can detect a division that costs roughly a quarter of the coverage. At the first position of this library, 60 of the 294 blocks form 30 pairs that are stereoisomers, indistinguishable under the achiral fingerprint used here and differing severalfold in in-class rate; that is a fifth of the catalogue carrying most of the binders, invisible to the test by construction.

**Table 7:** A label-free check, run on structures alone before any compound is assayed, against the coverage each design later achieved. The domain classifier separates audit compounds from batch compounds, which is the pair the predicted counts require; one half means they cannot be told apart. The block shares are the fraction of batch molecules whose building block at each position also occurs in the audit set.

| | domain | block share at position | | | |
|---|---|---|---|---|---|
| split design | classifier | 1 | 2 | 3 | cov. |
| random assignment | 0.498 | 1.00 | 1.00 | 1.00 | 98% |
| withhold pos. 3 | 0.985 | 1.00 | 1.00 | 0.00 | 48% |
| withhold pos. 2 | 0.989 | 1.00 | 0.00 | 1.00 | 37% |
| withhold pos. 1 | 0.997 | 0.00 | 1.00 | 1.00 | 35% |

## 3.11 Calibrating against a cheaper label

The audit sections asked which compounds a round must measure. The same assumption can fail along the other axis, in what is measured on them: the quantity a campaign cares about is an affinity, and what it can afford at scale is a proxy, such as a selection count or a single-shot readout. Thresholds are then set against the proxy while the statement the team wants concerns the affinity. The compounds may be a single random sample throughout, and they are in what follows, so nothing about the population is in question; it is the labelled pair that is no longer exchangeable, because the calibration set carries the proxy's label where the audit and the batch carry the campaign's. What that costs is not a question about how often the two labels agree.

One case is free, and for the reason that makes the construction rank-based. No part of the rule reads the numerical value of a score: a threshold is an order statistic of the calibration scores of one class, a prediction set follows from comparisons against it, and every count in the outcome table is a function of ranks. So the table is unchanged by any strictly increasing transformation of the score, which may therefore be a potency, a logit or an arbitrary monotone rescaling rather than a probability; we verify this on the library screen, where three such scales give tables agreeing cell for cell (ESI Section S15). The same argument applies to

the label. If the proxy is an increasing function of the affinity, then being above a proxy cut-off and being above the corresponding affinity cut-off are the same event, so the two labels coincide and nothing in the analysis moves.

Real proxies are not increasing functions of anything. What governs the shortfall is local, and follows from the construction: the threshold sits at a fixed quantile of the score distribution conditioned on the *proxy's* label, whereas the coverage achieved is the score distribution conditioned on the *true* label evaluated at that same threshold. The coverage achieved therefore decomposes into three terms,

$$\text{achieved} \;=\; (1-\alpha) \;+\; \big(\ell - (1-\alpha)\big) \;+\; \Delta, \qquad (1)$$

in which $\ell = F(\tau \mid \text{proxy out-class})$ is the realized level of the order statistic the threshold sits at and $\Delta = F(\tau \mid \text{true out-class}) - \ell$ is what the proxy costs. The middle term is the overshoot the batch correction introduces on purpose, and it is not small: the control below, whose proxy is the truth, measures $\ell = 0.924$ against a request of 0.90. Only the third term involves the proxy, and no part of it counts how many compounds the two labels disagree about. Measuring $\Delta$ against the control rather than against the nominal level is what the agreement quoted below tests; the decomposition itself is an accounting identity. Disagreement costs coverage only insofar as it is structured with respect to the score.[1]

Three carbonic anhydrase isoforms let this be separated, because selection counts and measured inhibition constants are published for the same targets[65,66] (Table 8). No molecule in that release carries both, so the two tables cannot be joined and no real selection count can be tested against an affinity here; the proxy is instead constructed from the affinities, which is what allows the structure of its errors to be set deliberately while the compounds and the score are held fixed. What follows is therefore a demonstration of the mechanism rather than a measurement of any particular assay's fitness as a proxy. Varying only the calibration label, we compare a strictly increasing relabelling of the affinity at matched prevalence, which is the control, against two proxies that relabel the same fraction of each class: one choosing those compounds uniformly at random, the other choosing the ones the score ranks most extremely against their class. The two differ from the truth on identically many compounds in each class, so their error rates, their ranking of the truth and the prevalence they imply are all equal; they differ only in where their mistakes sit in the ranking.

They behave nothing alike. At a fifth of each class relabelled, the random proxy left class-conditional coverage at 0.933 and 0.938 against a request of 0.90, falling short in 18 of 240 class-repeats, while the score-structured proxy gave 0.721 and 0.721 and fell short in 240 of 240, at every isoform and both cut-offs. Equation 1 accounts for this, and measuring both of its sides in the same run agrees to within 0.004 on average and 0.011 at worst across all 36 conditions: that is a consistency check on the accounting rather than a prediction, and its value is that it locates the quantity, not that it is accurate. The audit-set intervals, meanwhile, covered 95.2% of consequence lines over every condition, against their nominal 95%, and are within Monte Carlo error across the table. They ask only that the audit and the batch be exchangeable, not that the rule be well calibrated, so they price whatever rule they are handed; a campaign that calibrates on a proxy keeps its predicted counts and loses its coverage guarantee, and only the audit tells it so.

[1] Which is why a global summary of agreement, or a correlation between the proxy and the affinity, is the wrong thing to require of a proxy, and why comonotonicity is the right thing but is needed only locally: what Eq. 1 asks is that an isotonic regression[62,63] of the affinity on the proxy be increasing over the narrow score region where the thresholds fall, which the rule itself identifies as the region in which it declines to commit to a single label. Away from that region a proxy may misorder compounds freely. Fitting such a mapping needs compounds measured both ways, which a campaign can buy on a small sample; inference that corrects a cheap proxy using exactly such a small gold-standard sample is the subject of prediction-powered inference,[64] which repairs the bias where the account here instead characterizes when there is none to repair.

**Table 8:** Calibrating against a proxy label on three carbonic anhydrase isoforms, at affinity cut-offs of 7 and 8, 20 repeats each. The two proxies relabel the same fraction of each class and so agree with the truth equally well by any label-only summary; they differ in whether their mistakes are placed at random or where the score ranks compounds most extremely; the rows below name that placement. Coverage is class-conditional against a request of 0.90, pooled over isoforms and cut-offs; shortfall counts are given in the text rather than as a column.

| | relabelled | agree | coverage | | audit |
|---|---|---|---|---|---|
| proxy label | share | w/ truth | out | in | intervals |
| increasing in affinity | 0.00 | 1.000 | 0.924 | 0.923 | 96% |
| at random | 0.05 | 0.950 | 0.929 | 0.933 | 95% |
| by score | 0.05 | 0.950 | 0.873 | 0.872 | 95% |
| at random | 0.10 | 0.900 | 0.932 | 0.936 | 96% |
| by score | 0.10 | 0.900 | 0.822 | 0.824 | 94% |
| at random | 0.20 | 0.800 | 0.933 | 0.938 | 95% |
| by score | 0.20 | 0.800 | 0.721 | 0.721 | 95% |

That the requirement is local can be made exact, and in terms a campaign already uses. A team picks a quantile above which it will make compounds, and the rule here is of that shape, with one cut per class: at a request of $1-\alpha$ the out-class cut sits near the $1-\alpha$ quantile of the out-class scores and the in-class cut near the $\alpha$ quantile of the in-class scores, measured at 0.93 and 0.07 in these runs. Holding the relabelled share at a tenth of each class, so that the confusion matrix is the same in every condition, and moving only where in the within-class ranking those compounds are taken from, the damage follows the cuts and separates by class. Mistakes placed at the out-class cut displaced the out-class distributions by 0.036 and the in-class ones by 0.007; placed at the in-class cut the figures exchange, 0.007 and 0.036; placed anywhere between the two they left both at 0.007 to 0.009, which is the sampling floor. Coverage shortfalls track this, 54 of 120 class-repeats at a class's own cut against 6 to 19 elsewhere, and the audit intervals again covered 96.4% throughout. A proxy must therefore order compounds correctly in a narrow window around the quantile the campaign has chosen, and is free to misorder them everywhere else: the requirement attaches to a number the team states, not one it has to discover.

## 3.12 What the audit must sample

The audit describes a future batch drawn from the same library as the audit itself. In simulation, when compounds share scaffold-level shifts of their predicted probability, the intervals held when the audit sampled the scaffolds being screened: coverage was 95.4–97.9% (Fig. 7, left). Auditing other scaffolds does not substitute. When the future batch

came from new scaffolds with a scaffold effect of one standard deviation on the logit scale, coverage fell to 71% for false leads, 83% for missed in-class compounds and 88% for deferrals; with 1.5 standard deviations it fell to 62–75%.

**Tox21 scaffolds.** On Tox21 scaffold splits, calibration on other scaffolds broke the coverage guarantee: per-class coverage fell short in 28% of splits for out-class compounds and 18% for in-class compounds, two to three times the exact-law expectation (22.8 and 18.2 of 240 splits). An audit set from the screened scaffolds still reported the consequences correctly (Table 2). Calibrating instead on a random pilot sample of the screened scaffolds restored the coverage guarantee: per-class violations then followed the exact law (ESI Table S8).

**Solubility.** We also calibrated on all calibration compounds (about 1390) and then screened lipophilic compounds. Coverage of the soluble class among lipophilic audit compounds averaged 72% and fell below the requested 90% in all ten splits. Calibrating on the lipophilic compounds only gave 94% (below 90% in one split of ten, as the request allows), at the price of more measurements: 462 rather than 209 per 1000.

**How large a pilot must be.** In the Tox21 pilots, violations for the in-class compounds were nonetheless frequent, 29%, 13% and 12% for pilots of 200, 500 and 1000 compounds, against 26%, 12% and 9% expected, because many pilots held too few in-class compounds for the request to be attainable. At $\alpha = \delta = 0.10$ at least 22–23 calibration compounds of a class are needed, whatever the batch size; 70%, 41% and 11% of the pilots had fewer (median 13, 29 and 54 in-class compounds). A pilot of $22/\pi$ random compounds at prevalence $\pi$ has an expected in-class count of exactly 22, so it leaves the request unattainable in about half of draws; the pilots run here were infeasible in 70%, 41% and 11% of draws at 200, 500 and 1000 compounds. Sizing for a stated chance of feasibility rather than for the expectation, about $28/\pi$ reaches 22 in-class compounds nine times in ten, which is 980 for NR-PPAR-$\gamma$ against the 770 the expectation asks for. At that minimum the request is met only by the most conservative threshold; each further labelled compound then buys single-label calls.

**Using the calibration set as its own audit.** When labels are too scarce for a separate audit, the calibration set can partly serve as its own, and intervals built that way covered the future count of every cell in 97.7–100% of simulation replicates, 98.7% of realized counts on Tox21 random splits and 99.7% on the electrolyte screen (ESI Section S4). They are not a substitute, for two reasons the paper has already given. At a fixed budget of 1000 labelled compounds they are wider in every cell than splitting that budget between calibration and an audit — 21–28% for false leads and missed in-class compounds, 31–39% for confident hits and correct rejections and about twice as wide for deferrals (ESI Table S2) — while the larger calibration set buys almost nothing in return, moving the quantile the thresholds use only from 0.938 to 0.936 for a batch of 96. And being the calibration set, they inherit whatever mismatch it carries: calibrating on other scaffolds cut their coverage to 74% and on other salts to 33%. The audit set is the only one of the two that can be drawn from the chemistry the batch will contain.

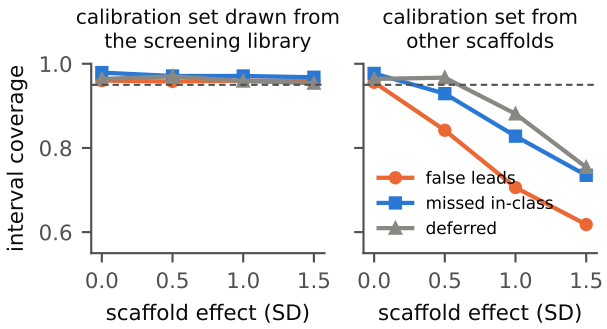



**Figure 7:** Coverage of 95% audit intervals when compounds are clustered in 60 scaffolds that share a shift of the predicted probability (3000 replicates per point). Left: calibration, audit and future compounds drawn from the same scaffolds, as with samples from the screening library. Right: future compounds from new scaffolds.

# 4 Discussion

The audit turns a conformal classifier into a forecast: before a batch is screened, it predicts how many hits, false leads, misses and measurements the batch will contain, with intervals that hold at a stated probability. The two class-wise coverage requests move a conformal classifier through the decisions available to a screening team, and for every setting the consequences are counts in one outcome table: compounds advanced, discarded and deferred, and how many of those calls are wrong. Linked to these counts by the audit, coverage becomes a coordinate a team can steer by. An objective, usually with guarantees on what a campaign will consume and deliver that hold with a stated probability, decides where to steer. The coverage plane is where such scenarios are explored, and an audit set from the library to be screened supplies the consequences, with finite-sample valid intervals that remain valid after the team has chosen a setting by looking at them. On Tox21, AqSolDB and CALiSol-23 the intervals held at their nominal level, also when the calibration came from other scaffolds or other salts, the audit having come from the screened compounds. Each of those checks divides one labelled pool at random, however, so they verify the construction rather than establishing that it survives a separately assembled batch, and the two divisions that do test that — by provenance and by building block — break it.

**Using the method in a screen.** The results suggest the following workflow:

1. Label a random sample of the library to be screened and use it as the calibration set. It must contain at least 22–23 compounds of the rarer class for a request at $\alpha = \delta = 0.10$ to be attainable, and more for a useful one: keep labelling randomly chosen compounds: $22/\pi$ at prevalence $\pi$ makes the request attainable only about half the time, and about $28/\pi$ reaches it nine times in ten. Choosing compounds by their score to find the rare class faster breaks exchangeability unless the selection

is accounted for.[67]

2. Label a second random sample as the audit set. Its size sets the width of the intervals: for an outcome with frequency $p$ in a batch of $m$, $n$ audit compounds give a 95% interval of roughly $\pm 2m\sqrt{p(1-p)(1/n+1/m)}$, about $\pm 35$ per 1000 for $p = 0.1$ and $n = 400$ (ESI Section S3). Choose $n$ from the width the decision needs, not from validity, which any audit size delivers. Read the floor first, since no audit narrows an interval below $\pm 2\sqrt{m\,p(1-p)}$: accepting $r$ times that floor needs $n = m/(r^2-1)$, so an audit the size of the batch is the natural place to stop and an audit much larger than the batch buys almost nothing. Batch size decides what this costs. For a 96-well plate a few hundred audit compounds already sit within a fifth of the floor, and buying more is close to wasted; for a batch of 1000 the same audit sits more than twice the floor, and closing that gap means labelling on the scale of the batch itself. With class probabilities and few labels, the calibration set can serve instead, at a cost in width that Section 3.12 quantifies at the same labelling budget.

3. Draw the audit from the chemistry the next batch will contain. Where a batch introduces building blocks, scaffolds or series the audit does not hold, matching its composition is not enough: an audit carrying the same proportion of new chemistry drawn from different blocks performed no better than one ignoring the shift, while an audit sampling the batch's own blocks recovered the nominal level (Section 3.8). If part of the next batch is novel in this sense, a pilot of that chemistry is the indicated step, and on the electrolyte compilation it restores most of what a provenance boundary costs even when the pilot is divided from the batch non-randomly.

4. Before committing the batch, check that the calibration set, the audit set and the batch cannot be told apart. Training a classifier to separate calibration from batch compounds on structures alone needs no measurements and can be run on the proposed batch: an area under the curve near one half is consistent with exchangeability, and anything well above it means the intervals do not apply (Section 3.10). Comparing how many of the batch's scaffolds, series or building blocks occur in the audit says which part of the chemistry is responsible.

5. Sweep the coverage requests and audit each setting: intervals for missed in-class compounds, false leads and deferrals in the planned batch, with a Bonferroni correction over the sweep.

6. State a scenario, an objective and the limits that must hold with probability at least $1-\varepsilon$ (assay capacity, budget, minimum yield), and choose the best setting among those whose audit bounds satisfy the limits. If the objective and limits are monotone in the outcomes, the table of Pareto settings is all the team needs to see.

7. Screen, and compare the realized counts with the predicted intervals as a check on exchangeability.

**The audit decides which questions can be asked.** Calibration and audit sets are usually treated as machinery, chosen for size and then forgotten. They are better understood as the statement of what the guarantees are about, because a sample licenses claims about the population it was drawn from and about nothing else. How the audit is built therefore fixes, before any model is fitted, which questions a campaign can answer, and three choices in building it each foreclose something. The population fixes the domain: an audit drawn from the top of a ranking prices further batches drawn from the top of that same ranking, and says nothing about compounds further down, which is where a campaign goes once the top is exhausted. The unit fixes what a count means: an audit sampled one compound per series estimates the rate at which series are real and prices a batch of new series, not a batch of analogues of a few confirmed ones, whose effective size is the number of series it contains. The chemistry fixes the reach: an audit holding only building blocks already screened prices batches built from those blocks, and carrying the same proportion of new chemistry as the batch without sampling that chemistry is not a substitute (Section 3.8). The label fixes the claim: what must be exchangeable is each candidate together with its recorded label, so a campaign that sets its thresholds against a cheap readout has made a statement about that readout and not about the quantity it cares about, however interchangeable the compounds are (Section 3.11). The score belongs to the same declaration, since thresholds and the sets they define move when the model is refitted, so an audit describes the deployment it was measured under.

Reading the four together makes the condition less forbidding than it first appears, because exchangeability is sufficient but more than is needed. A coverage request concerns one quantile, and what the request actually buys is an agreement between calibration and batch at the cut it induces; a discrepancy elsewhere in the ranking is free. That is why a proxy whose errors fall away from the cut costs nothing while one whose errors fall on it is ruinous at the same error rate, and it suggests that the familiar violations might be graded the same way, by where in the ranking a population differs rather than by whether it differs at all. We have tested this only on the label, where the structure of the discrepancy can be set deliberately; doing the same for a change of chemistry would ask how much of the loss under a block holdout is attributable to the neighbourhood of the thresholds, which the designs here do not separate.

Two consequences follow for practice. The question should be settled before the audit is built, because an audit assembled for one question usually cannot be repurposed for another; a campaign that discovers halfway through that it needs to speak about a different population has to measure again. And because an audit is also an option on the questions a campaign may ask later, it is worth spending part of it on compounds a project would not otherwise measure, lower-ranked compounds or a few members of each of several series, so that when the obvious batch is exhausted the next question can still be answered. Diversity in an audit is not a statistical nicety but the purchase of future questions.

**Campaigns run batch by batch.** Automated and self-driving laboratories synthesize and characterize in parallel, so their campaigns are sequences of batches, each with a fixed capacity, and the framework applies to each batch as it stands. Three features of that setting make counts the right object and make an audit cheap. The constraints are phys-

ical and integer: a plate has 96 or 384 wells, a run has a reagent budget, and a round must return enough confirmed hits to justify the next. None of those is a constraint on a rate, and a scheduler cannot act on one — it needs the probability that this plate exceeds capacity, which is what a count interval states and what a risk bound does not. The batches are small and frequent, which is the regime in which the draw of the batch dominates the uncertainty, so the gap between a rate and a count is widest exactly here. And the audit is affordable because the unit of cost is the round: reserving a slice of a committed plate spends candidate slots, not an extra round. Batch campaigns also tend to pursue different goals from a search for a single optimum, though batch Bayesian optimization has long addressed the parallel setting directly. Because a batch is evaluated in parallel, its members are chosen to be diverse,[68–70] and the campaign often seeks coverage of a chemical or physical design space: mapping a landscape, as in space-filling designs and closed-loop phase mapping,[71,72] contracting onto promising regions,[73] or finding a set of optima such as a Pareto front.[74] In each of these modes a batch is a selection from a scored library under a fixed capacity, and its consequences are what the audit predicts. The batch size $m$ is the plate or run size, and the scenario constraints are the capacity of the platform, the reagent or instrument budget per run and the minimum yield needed to proceed. What the outcome table accounts for is which prediction set a candidate receives, not how the batch it belongs to was assembled. A planner that fills a plate with the highest-ranked candidates does not draw a random sample of the library, so an audit reserved from the library does not price that plate — which Section 3.9 measures rather than assumes, and which fails for a reason a team can see without any model: the plate is several times richer than the collection it was chosen from. The repair is to sample the plate and not the library, and it is exact, because a random slice of a committed batch is exchangeable with the rest of that batch however the batch was chosen. A campaign that adapts therefore reserves a random slice of every committed round, and the guarantee attaches to the remainder of that round. Methods for data chosen by the model are the relevant prior art here rather than a complementary aside.[67,75,76] When the model is retrained between batches, the thresholds change, so the calibration, the audit and the map are redone for the new deployment. Two practices follow. First, reserving a small random audit slice in every batch, measured whatever the rule decided, checks continually that the screened library still matches the calibration data; it is the acceptance sample of quality control applied to each lot. Second, the guarantee holds per batch with probability $1-\varepsilon$; a statement over all $K$ batches of a campaign needs level $\varepsilon/K$ for each. Because the audit bounds are stated in counts, they can be checked automatically, and the Pareto table is a compact menu from which a planner, including an AI agent, can choose a setting without a cost ratio being stated; we do not test such a planner here. What the campaign experiment adds is that the rule such a planner must follow is short and does not depend on its acquisition function: reserve a random slice of each committed round, price the remainder with it, and draw a fresh slice next round. The first clause holds because a random division of a committed batch is exchangeable with its remainder however the batch was chosen, so the planner's selection drops out of the argument entirely and no correction for it is needed; the third because one round at a sub-percent hit rate removes most of what was there to find, so an audit prices the round it came from and not its successor. A platform that follows those two clauses has finite-sample bounds on every quantity its scheduler is constrained by, for the plate it is about to commit.

**Why parallel changes the economics.** Automated campaigns are often pictured as sequential loops that adapt after every experiment, and a loop of that kind corrects its own errors: it proposes, measures, and the surprise moves the model, whether or not the uncertainty it reported was calibrated. Laboratories do not work that way, because the cost of an experiment is dominated by the round rather than by the candidate. A plate, a synthesis run or an instrument cycle costs nearly the same whether it is full or half empty, so cost rises in steps at each new round and barely at all within one. Three consequences follow, and they run against the intuition that smaller, more frequent batches are the way to keep a campaign honest.

The first is that an audit is cheap. When the round is the unit of cost, reserving part of a committed batch for a random sample of the library costs a few candidate slots rather than an extra round, so the measurements that price the next batch are close to free in money and are what the guarantees are bought with. The second is that correction is bought by composition rather than by cadence. Running more rounds to adapt sooner pays the fixed cost repeatedly, whereas reserving slots and spending the confidence level across a declared number of looks costs a fraction of one batch (Section 3.3). The third runs opposite to the sequential picture: the larger the batch that must be committed before anything is learned, the less opportunity there is to adapt within it, and the more of the decision rests on what was predicted beforehand. Parallelism and predicted consequences are therefore complements rather than alternatives. A laboratory that could run one cheap experiment at a time would have little use for an interval on a batch; a laboratory that must commit several hundred wells at once has the most use for one. On that reading a committed batch is an allocation under fixed capacity, divided between the candidates a setting advances, the random sample that prices the following round, and, where the next round will reach into chemistry the audit does not hold, a pilot of that chemistry (Section 3.8).

**Limitations: what breaks exchangeability.** All guarantees rest on exchangeability between the calibration, audit and screened candidates, each taken together with its recorded label. A change of chemical series, scaffold or assay breaks it, as the scaffold results show, and so does calibrating against a label other than the one the claim concerns, which breaks it without touching the composition of any set. So does a batch selected with the model, as in active learning, unless the selection is part of the design, and so does a library proposed by a generative model or extended into new chemistry, as the new-salt screen shows;[67,76] methods for shifted data are complementary.

**Correlated candidates.** Compounds that share scaffolds are correlated, so the effective size of a pilot or audit sample is closer to the number of series sampled than to the number of compounds, as the account of what an audit decides

sets out above. We can now put numbers to it. Decomposing the variance of the label within and between clusters gives an intracluster correlation of 0.60 for the electrolytes at the level of the salt–solvent system, over the 138 systems that carry one, 10.8 formulations each, hence a design effect of 6.9: the 1495 room-temperature formulations are worth about 220 independent ones. For the library at the level of the first-position block the correlation is 0.35 with 680 molecules per block, a design effect of 238, so the 200 000 molecules analysed are worth about 840. What this does and does not touch is worth separating. It does not touch the audit intervals: the hypergeometric construction conditions on the total count in the audit and the batch together, which absorbs the between-cluster variance, so under random assignment both the coverage and the width of an interval are invariant to the intracluster correlation. Nor does it touch the audit-size rule, since both variance terms inflate together and the ratio $r$ cancels the design effect. Where it bites is any statement that treats a count of candidates as a count of independent observations: the reach of a pilot, the number of clusters a sweep has actually explored, and the confidence one may place in a coverage figure estimated from correlated splits. The attainability rule for a pilot is a simple-random-sample binomial result; a pilot of new chemistry is drawn by whole blocks, so its in-class count is overdispersed, and a sample-size rule inflates by the design effect itself rather than by its square root.[57,58] The pools used here are diverse enough for this to be immaterial, with 328 similarity clusters among 384 audited solubility compounds and over 1200 among the Tox21 compounds. It governs libraries built combinatorially or around a single target, and the DNA-encoded library is one: its random-assignment arm, which we report as the design under which the intervals hold, draws calibration, audit and batch from molecules that differ by single building blocks, so its effective sample size is smaller than its molecule count and the agreement we find there should be read as a check on the construction rather than as a measure of how much such a sample is worth.

**A structural check does not substitute for a pilot.** We found no relation between how well an interval covered and how similar the held-out building blocks were to those the audit held; blocks indistinguishable by fingerprint gave the lowest coverage of all, being stereoisomers whose in-class rates differ severalfold, so similarity measured on an achiral representation cannot certify a batch that differs stereochemically from its audit. The same limit bounds the pre-flight test, which inherits whatever the representation cannot see: it separated every block holdout here and the two preparations that fail, but a shift carried by features the fingerprints omit would pass it, and the provenance division of Section 3.7 is exactly such a shift. Its verdicts are therefore asymmetric, refuting an arrangement rather than certifying one, and a richer representation would widen what it can refute without ever making a pass conclusive.

**The price of simultaneity, and the reach of the sweep.** Each interval covers one outcome of one setting; statements about many outcomes or settings need the Bonferroni correction, which widens intervals; with 144 requests and four bounded outcomes, the joint guarantee cost about 110 measurements per 1000 at an audit of 384 compounds. Larger audits lower this price, and bounds that exploit the monotone structure of the sweep could lower it further; an envelope valid simultaneously over all thresholds is the natural instrument, and one now exists for the false discovery proportion of a conformal selection.[31] The sweep reaches every two-threshold rule only with class probabilities, and the cost-optimal rule is a two-threshold rule only if the probability of being in-class increases with the score. Cost ratios are rarely known precisely, so we report the range of ratios for which a setting is cheapest.

**The datasets and their labels.** Tox21 labels are missing for 7.6–26.0% of the 7592 compounds depending on the endpoint, and the missingness is not random: the 2926 compounds with no missing endpoint are active in 2.7% of their labelled cells, the 3188 missing one to three in 8.9%, and the 1478 missing four or more in 23.2%. The blanks are the inconclusive calls of the original screen, which fall disproportionately on cytotoxic and interfering compounds — the kind a real primary screen contains and a model scores high. Every Tox21 audit set here is therefore a random sample of the *labelled* subpopulation rather than of the library, at a lower in-class rate than the library carries. That is an instance of this paper's own thesis rather than an exception to it, and it means the Tox21 results should be read as verification of the construction on a clean pool rather than as evidence about a screen of the whole collection. AqSolDB combines measurements from many sources, with a typical spread of 0.5–0.7 log units between sources;[54] labels near the cut-off at $\log S = -4$ are therefore noisy, and at that spread the expected label-flip rate at the cut is 8–9% among the lipophilic compounds screened, which is the size of the coverage requests being swept; Section 3.4 measures what that costs, and the scenario is the part it costs the most; the curation rule also removed the most discordant duplicates, which a real batch would contain. CALiSol-23 likewise pools 27 sources, and several formulations were measured at many temperatures or concentrations, so neighbouring formulations are correlated. Its label also carries the provenance of the measurement rather than only the chemistry: 90% of room-temperature formulations have a single measurement in the 293–303 K window, and the temperature of that measurement is bimodal, 365 below 296 K and 917 above 300 K against 200 near 298 K, while conductivity rises by a median $0.136\,\mathrm{mS\,cm^{-1}\,K^{-1}}$, so the window spans 17% of the $8\,\mathrm{mS\,cm^{-1}}$ cut and a quarter of the formulations sit within 20% of it. Label temperature is moreover confounded with salt and with source, so the salt holdouts change the measurand as well as the chemistry. Two entries of the salt column, LiTFSI and $\mathrm{LiN(CF_3SO_2)_2}$, are the same compound under two names, 27 and 9 room-temperature rows whose mean label temperatures differ by five degrees; they were encoded separately and can therefore be split across a salt holdout. Both should be repaired before the electrolyte results are relied on: the first by interpolating each formulation to a common temperature, the second by merging the names. The guarantees refer to the labels as recorded.

**Scope.** We treat binary endpoints only, and one model and representation throughout. The pre-flight test, the stereoisomer blindness and the behaviour of the ranking metric at subpercent prevalence are all properties of that one fingerprint,

so the design instruction of Section 3.10 is validated on a representation this section shows to be blind to a fifth of the first-position block catalogue; whether that instruction is advice or an artefact of the representation is not settled here.

# 5 Conclusions

The coverage level of a conformal classifier is a coordinate for navigating screening decisions, and the audited consequences, counted in compounds and bounded with a stated probability, are what a team chooses by under its own objective and guarantees. Each setting sorts a future batch into one table of prediction set by true label, and every screening quantity is a sum of its cells. A labelled audit set from the library to be screened predicts all of these counts for the next batch, with intervals that hold with a specified probability in finite samples, under exchangeability of the audit with the batch alone and after a setting is chosen. The same request led to very different consequences across twelve Tox21 endpoints, and one model offered more than a hundred distinct trade-offs for a solubility screen and for an electrolyte screen. When the calibration did not match the screened population, as for a new lithium salt, the audit still bounded the consequences correctly — though that check, like every positive coverage result here, divides one labelled pool at random, and the two divisions that test the intervals against a genuinely separate batch, by provenance and by building block, break them. In automated campaigns, which run in parallel batches, the same construction applies to each batch, and the bounds become capacity constraints a planner can check, provided the slice it reserves comes from the committed round and not from the library: a plate chosen from the top of a ranking was 38 times richer than the pool it came from, and an audit of that pool held nothing in-class to measure. One round also depletes what remains, so the slice is re-drawn each round rather than amortized across the campaign. Inverted, the same intervals size a round: the batch an audited count implies for a target number of hits is larger than dividing the target by the rate suggests, by a factor that grows as the rate falls, and the audit that pins the rate down is the measurement with the largest effect on what the round costs. What the audit must contain is decided by the batch it prices, since an audit that reproduces a batch's composition without sampling its chemistry prices nothing; buying a pilot of the screened population does repair it, and does so under divisions that are not random and could have failed. With an audit built this way, a team sees those consequences before the campaign starts.

# Author contributions

P.H.Z. conceived the study, developed the methodology, performed the analysis and wrote the manuscript.

# Conflicts of interest

There are no conflicts to declare.

# Data availability

Tox21 is available through MoleculeNet, AqSolDB through its dataset reference[54] and CALiSol-23 from the DTU data repository (DOI 10.11583/DTU.c.6929599).[55] The carbonic anhydrase selection counts and affinities are from the CA-DEL release[65] (Zenodo record 17656024, DOI 10.5281/zenodo.17656024, CC BY 4.0), whose affinity values derive from ChEMBL[66] and carry CC BY-SA 3.0. The DNA-encoded library is BELKA,[56] available from Leash Biosciences through the Polaris record `leash-bio/belka-v1` under CC BY-NC-SA 4.0; the million-molecule sample used here is reproduced by the seeded digest rule given in ESI Section S14. The scripts that reproduce every table and figure are provided with the ESI, together with the processed data, split indices and stored model scores. They implement the finite-sample threshold corrections, randomized tie-breaking, outcome tables, prediction intervals for future batches from an audit set or from the calibration set alone, coverage-request sweeps with corrections for selection, and the cost-ratio analysis. They are released under the MIT licence at `https://github.com/phzwart/conformal-screening`. [Tagged version and archived Zenodo DOI, on acceptance.]

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# References

[1] Milad Abolhasani and Eugenia Kumacheva. The rise of self-driving labs in chemical and materials sciences. *Nat. Synth.*, 2:483–492, 2023. doi: 10.1038/s44160-022-00231-0.

[2] Gary Tom, Stefan P. Schmid, Sterling G. Baird, Yang Cao, Kourosh Darvish, Han Hao, Stanley Lo, Sergio Pablo-García, Ella M. Rajaonson, Marta Skreta, Naruki Yoshikawa, Samantha Corapi, Gun Deniz Akkoc, Felix Strieth-Kalthoff, Martin Seifrid, and Alán Aspuru-Guzik. Self-driving laboratories for chemistry and materials science. *Chem. Rev.*, 124(16):9633–9732, 2024. doi: 10.1021/acs.chemrev.4c00055.

[3] Vladimir Vovk, Alex Gammerman, and Glenn Shafer. *Algorithmic Learning in a Random World*. Springer, New York, 2005. doi: 10.1007/b106715.

[4] Anastasios N. Angelopoulos and Stephen Bates. Conformal prediction: A gentle introduction. *Foundations and Trends in Machine Learning*, 16(4):494–591, 2023. doi: 10.1561/2200000101.

[5] Ulf Norinder, Lars Carlsson, Scott Boyer, and Martin Eklund. Introducing conformal prediction in predictive modeling. A

transparent and flexible alternative to applicability domain determination. *Journal of Chemical Information and Modeling*, 54(6):1596–1603, 2014.

[6] Isidro Cortés-Ciriano and Andreas Bender. Concepts and applications of conformal prediction in computational drug discovery. *arXiv preprint arXiv:1908.03569*, 2019.

[7] Siddhartha Laghuvarapu, Zhen Lin, and Jimeng Sun. CoDrug: conformal drug property prediction with density estimation under covariate shift. In *Advances in Neural Information Processing Systems*, volume 36, 2023.

[8] Stephen Bates, Anastasios Angelopoulos, Lihua Lei, Jitendra Malik, and Michael I. Jordan. Distribution-free, risk-controlling prediction sets. *Journal of the ACM*, 68(6), September 2021. ISSN 0004-5411. doi: 10.1145/3478535. URL `https://doi.org/10.1145/3478535`.

[9] Anastasios N. Angelopoulos, Stephen Bates, Adam Fisch, Lihua Lei, and Tal Schuster. Conformal risk control. In *International Conference on Learning Representations*, 2024.

[10] Anastasios N. Angelopoulos, Stephen Bates, Emmanuel J. Candès, Michael I. Jordan, and Lihua Lei. Learn then test: calibrating predictive algorithms to achieve risk control. *arXiv preprint arXiv:2110.01052*, 2021.

[11] Lewis H. Mervin, Simon Johansson, Elizaveta Semenova, Kathryn A. Giblin, and Ola Engkvist. Uncertainty quantification in drug design. *Drug Discovery Today*, 26(2):474–489, 2021. doi: 10.1016/j.drudis.2020.11.027.

[12] A. R. Thatcher. Relationships between Bayesian and confidence limits for predictions. *Journal of the Royal Statistical Society: Series B (Methodological)*, 26(2):176–192, 1964. doi: 10.1111/j.2517-6161.1964.tb00551.x.

[13] William Q. Meeker, Gerald J. Hahn, and Luis A. Escobar. *Statistical Intervals: A Guide for Practitioners and Researchers*. Wiley, Hoboken, NJ, 2nd edition, 2017.

[14] Ruben Buendia, Thierry Kogej, Ola Engkvist, Lars Carlsson, Henrik Linusson, Ulf Johansson, Paolo Toccaceli, and Ernst Ahlberg. Accurate hit estimation for iterative screening using Venn–ABERS predictors. *J. Chem. Inf. Model.*, 59(3):1230–1237, 2019. doi: 10.1021/acs.jcim.8b00724.

[15] David Lopez-Paz and Maxime Oquab. Revisiting classifier two-sample tests. In *International Conference on Learning Representations*, 2017.

[16] Robert P. Sheridan, Bradley P. Feuston, Vladimir N. Maiorov, and Simon K. Kearsley. Similarity to molecules in the training set is a good discriminator for prediction accuracy in QSAR. *Journal of Chemical Information and Computer Sciences*, 44 (6):1912–1928, 2004. doi: 10.1021/ci049782w.

[17] Ulf Norinder and Scott Boyer. Binary classification of imbalanced datasets using conformal prediction. *Journal of Molecular Graphics and Modelling*, 72:256–265, 2017.

[18] Jiangming Sun, Lars Carlsson, Ernst Ahlberg, Ulf Norinder, Ola Engkvist, and Hongming Chen. Applying Mondrian cross-conformal prediction to estimate prediction confidence on large imbalanced bioactivity data sets. *Journal of Chemical Information and Modeling*, 57(7):1591–1598, 2017.

[19] Fredrik Svensson, Ulf Norinder, and Andreas Bender. Modelling compound cytotoxicity using conformal prediction and PubChem HTS data. *Toxicology Research*, 6(1):73–80, 2017.

[20] Nicolas Bosc, Francis Atkinson, Eloy Felix, Anna Gaulton, Anne Hersey, and Andrew R. Leach. Large scale comparison of QSAR and conformal prediction methods and their applications in drug discovery. *Journal of Cheminformatics*, 11:4, 2019.

[21] Andrea Morger, Fredrik Svensson, Staffan Arvidsson McShane, Niharika Gauraha, Ulf Norinder, Ola Spjuth, and Andrea Volkamer. Assessing the calibration in toxicological in vitro models with conformal prediction. *Journal of Cheminformatics*, 13:35, 2021. doi: 10.1186/s13321-021-00511-5.

[22] Gökçe Geylan, Leonardo De Maria, Ola Engkvist, Florian David, and Ulf Norinder. A methodology to correctly assess the applicability domain of cell membrane permeability predictors for cyclic peptides. *Digital Discovery*, 3:1761–1775, 2024. doi: 10.1039/D4DD00056K.

[23] Staffan Arvidsson McShane, Ulf Norinder, Jonathan Alvarsson, Ernst Ahlberg, Lars Carlsson, and Ola Spjuth. CP-Sign: conformal prediction for cheminformatics modeling. *Journal of Cheminformatics*, 16:75, 2024. doi: 10.1186/s13321-024-00870-9.

[24] Ruben Buendia, Ola Engkvist, Lars Carlsson, Thierry Kogej, and Ernst Ahlberg. Venn–Abers predictors for improved compound iterative screening in drug discovery. *Proc. Mach. Learn. Res.*, 91:201–219, 2018.

[25] Fredrik Svensson, Avid M. Afzal, Ulf Norinder, and Andreas Bender. Maximizing gain in high-throughput screening using conformal prediction. *Journal of Cheminformatics*, 10:7, 2018. doi: 10.1186/s13321-018-0260-4.

[26] Vladimir Vovk. Conditional validity of inductive conformal predictors. *arXiv*, 2012. doi: 10.48550/arXiv.1209.2673. Extended version of ACML 2012 paper.

[27] Michelle Bian and Rina Foygel Barber. Training-conditional coverage for distribution-free predictive inference. *Electronic Journal of Statistics*, 17(2):2044–2066, 2023. doi: 10.1214/23-EJS2145.

[28] Paulo C. F. Marques. Universal distribution of the empirical coverage in split conformal prediction. *Statistics & Probability Letters*, 219:110350, 2025. ISSN 0167-7152. doi: 10.1016/j.spl.2024.110350. URL `https://www.sciencedirect.com/science/article/pii/S0167715224003195`.

[29] S. S. Wilks. Determination of sample sizes for setting tolerance limits. *The Annals of Mathematical Statistics*, 12(1): 91–96, 1941.

[30] Sangdon Park, Osbert Bastani, Nikolai Matni, and Insup Lee. PAC confidence sets for deep neural networks via calibrated prediction. In *International Conference on Learning Representations*, 2020.

[31] Ziang Song, Ying Jin, and Emmanuel J. Candès. Everywhere valid bounds on false discovery proportions in conformal inference. *arXiv preprint arXiv:2605.20726*, 2026.

[32] Jelle J. Goeman and Aldo Solari. Multiple testing for exploratory research. *Statistical Science*, 26(4):584–597, 2011. doi: 10.1214/11-STS356.

[33] Eugene Katsevich and Aaditya Ramdas. Simultaneous high-probability bounds on the false discovery proportion in structured, regression and online settings. *The Annals of Statistics*, 48(6):3465–3487, 2020. doi: 10.1214/19-AOS1938.

[34] Gilles Blanchard, Pierre Neuvial, and Etienne Roquain. Post hoc confidence bounds on false positives using reference families. *The Annals of Statistics*, 48(3):1281–1303, 2020. doi: 10.1214/19-AOS1847.

[35] Ying Jin and Emmanuel J. Candès. Selection by prediction with conformal p-values. *Journal of Machine Learning Research*, 24(244):1–41, 2023.

[36] Mauricio Sadinle, Jing Lei, and Larry Wasserman. Least ambiguous set-valued classifiers with bounded error levels. *Journal of the American Statistical Association*, 114(525):223–234, 2019.

[37] Jordan Lekeufack, Anastasios N. Angelopoulos, Andrea Bajcsy, Michael I. Jordan, and Jitendra Malik. Conformal decision theory: Safe autonomous decisions from imperfect predictions. In *2024 IEEE International Conference on Robotics and Automation (ICRA)*, pages 11668–11675, 2024.

[38] Shayan Kiyani, George Pappas, Aaron Roth, and Hamed Hassani. Decision theoretic foundations for conformal prediction: Optimal uncertainty quantification for risk-averse agents. In *Proceedings of the 42nd International Conference on Machine Learning*, volume 267 of *Proceedings of Machine Learning Research*, 2025.

[39] C. K. Chow. On optimum recognition error and reject tradeoff. *IEEE Transactions on Information Theory*, 16(1):41–46, 1970. doi: 10.1109/TIT.1970.1054406.

[40] Foster Provost and Tom Fawcett. Robust classification for imprecise environments. *Machine Learning*, 42(3):203–231, 2001.

[41] Tadeusz Pietraszek. Optimizing abstaining classifiers using ROC analysis. In *Proceedings of the 22nd International Conference on Machine Learning*, pages 665–672, 2005.

[42] Pascal Massart. The tight constant in the dvoretzky–kiefer–wolfowitz inequality. *Annals of Probability*, 18(3):1269–1283, 1990. doi: 10.1214/aop/1176990746.

[43] A. Charnes and W. W. Cooper. Chance-constrained programming. *Management Science*, 6(1):73–79, 1959.

[44] Philip B. Stark. Conservative statistical post-election audits. *The Annals of Applied Statistics*, 2(2):550–581, 2008. doi: 10.1214/08-AOAS161.

[45] Mark Lindeman and Philip B. Stark. A gentle introduction to risk-limiting audits. *IEEE Security & Privacy*, 10(5):42–49, 2012. doi: 10.1109/MSP.2012.56.

[46] Harold F. Dodge and Harry G. Romig. *Sampling Inspection Tables: Single and Double Sampling*. Wiley, New York, 1944.

[47] Tilmann Gneiting, Fadoua Balabdaoui, and Adrian E. Raftery. Probabilistic forecasts, calibration and sharpness. *J. R. Stat. Soc. B*, 69(2):243–268, 2007. doi: 10.1111/j.1467-9868.2007.00587.x.

[48] Greg Landrum. RDKit: Open-source cheminformatics. `https://www.rdkit.org`, 2025. [Online; accessed 2026-02-13].

[49] David Rogers and Mathew Hahn. Extended-connectivity fingerprints. *Journal of Chemical Information and Modeling*, 50 (5):742–754, 2010. doi: 10.1021/ci100050t.

[50] Liudmila Prokhorenkova, Gleb Gusev, Aleksandr Vorobev, Anna Veronika Dorogush, and Andrey Gulin. CatBoost: unbiased boosting with categorical features. In *Advances in Neural Information Processing Systems*, volume 31, pages 6638–6648, 2018.

[51] Ruili Huang, Menghang Xia, Dac-Trung Nguyen, Jianying Zhao, Srilatha Sakamuru, Tingjun Zhao, Maxine Na, Swati A. Shahane, Anastasia Rossoshek, and Anton Simeonov. The tox21 data challenge to build predictive models of nuclear receptor and stress response pathways. *Nature Biotechnology*, 34(8):828–837, 2016. doi: 10.1038/nbt.3659.

[52] Andreas Mayr, Günter Klambauer, Thomas Unterthiner, and Sepp Hochreiter. Deeptox: Toxicity prediction using deep learning. *Frontiers in Environmental Science*, 3:80, 2016. doi: 10.3389/fenvs.2015.00080.

[53] Guy W. Bemis and Mark A. Murcko. The properties of known drugs. 1. Molecular frameworks. *Journal of Medicinal Chemistry*, 39(15):2887–2893, 1996. doi: 10.1021/jm9602928.

[54] Murat Cihan Sorkun, Abhishek Khetan, and Süleyman Er. Aqsoldb: a curated reference set of aqueous solubility and 2d descriptors for a diverse set of compounds. *Scientific Data*, 6: 143, 2019. doi: 10.1038/s41597-019-0151-1. URL `https://www.nature.com/articles/s41597-019-0151-1`.

[55] Paolo de Blasio, Jonas Elsborg, Tejs Vegge, Eibar Flores, and Arghya Bhowmik. CALiSol-23: Experimental electrolyte conductivity data for various Li-salts and solvent combinations. *Scientific Data*, 11(1):750, 2024.

[56] Ian Quigley, Andrew Blevins, Nate Wilkinson, Maxwell Kleinsasser, Victor Shlepov, Vlad Vinogradov, and Jonathan Hsu. BELKA: the Big Encoded Library for Chemical Assessment. In *Advances in Neural Information Processing Systems 37, Competition Track*, 2024. URL `https://openreview.net/forum?id=zwppB4butE`. Dataset: Leash Biosciences; Polaris record `leash-bio/belka-v1`, CC BY-NC-SA 4.0.

[57] William G. Cochran. *Sampling Techniques*. John Wiley & Sons, New York, 3rd edition, 1977.

[58] Leslie Kish. *Survey Sampling*. John Wiley & Sons, New York, 1965.

[59] K. K. Gordon Lan and David L. DeMets. Discrete sequential boundaries for clinical trials. *Biometrika*, 70(3):659–663, 1983. doi: 10.1093/biomet/70.3.659.

[60] Steven R. Howard, Aaditya Ramdas, Jon McAuliffe, and Jasjeet Sekhon. Time-uniform, nonparametric, nonasymptotic confidence sequences. *The Annals of Statistics*, 49(2):1055–1080, 2021. doi: 10.1214/20-AOS1991.

[61] Aaditya Ramdas, Peter Grunwald, Vladimir Vovk, and Glenn Shafer. Game-theoretic statistics and safe anytime-valid inference. *Statistical Science*, 38(4):576–601, 2023. doi: 10.1214/23-STS894.

[62] Miriam Ayer, H. D. Brunk, G. M. Ewing, W. T. Reid, and Edward Silverman. An empirical distribution function for sampling with incomplete information. *The Annals of Mathematical Statistics*, 26(4):641–647, 1955. doi: 10.1214/aoms/1177728423.

[63] Bianca Zadrozny and Charles Elkan. Transforming classifier scores into accurate multiclass probability estimates. In *Proceedings of the Eighth ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, pages 694–699, 2002. doi: 10.1145/775047.775151.

[64] Anastasios N. Angelopoulos, Stephen Bates, Clara Fannjiang, Michael I. Jordan, and Tijana Zrnic. Prediction-powered inference. *Science*, 382(6671):669–674, 2023. doi: 10.1126/science.adi6000.

[65] Yuxuan He, Duanhua Cao, Xiaozhe Tan, Mingan Gao, Xiaojun Yao, Shuangjia Gu, and Pheng-Ann Heng. CA-DEL: an open multi-target, multi-modal benchmark for learning from DNA-encoded library screens, 2026. Data: Zenodo record 17656024, *CA_DEL_database*, CC BY 4.0.

[66] Barbara Zdrazil, Eloy Felix, Fiona Hunter, Emma J. Manners, James Blackshaw, Sybilla Corbett, Marleen de Veij, Harris Ioannidis, David Mendez Lopez, Juan F. Mosquera, Maria Paula Magarinos, Nicolas Bosc, Ricardo Arcila, Tevfik Kizilören, Anne Gaulton, A. Patrícia Bento, Melissa F. Adasme, Peter Monecke, Gregory A. Landrum, and Andrew R. Leach. The ChEMBL database in 2023: a drug discovery platform spanning multiple bioactivity data types and time periods. *Nucleic Acids Research*, 52(D1):D1180–D1192, 2024. doi: 10.1093/nar/gkad1004.

[67] Rina Foygel Barber, Emmanuel J. Candès, Aaditya Ramdas, and Ryan J. Tibshirani. Conformal prediction beyond exchangeability. *The Annals of Statistics*, 51(2):816–845, 2023.

[68] Javier González, Zhenwen Dai, Philipp Hennig, and Neil Lawrence. Batch Bayesian optimization via local penalization. In *Proceedings of the 19th International Conference on Artificial Intelligence and Statistics*, volume 51 of *Proceedings of Machine Learning Research*, pages 648–657, 2016.

[69] José Miguel Hernández-Lobato, James Requeima, Edward O. Pyzer-Knapp, and Alán Aspuru-Guzik. Parallel and distributed Thompson sampling for large-scale accelerated exploration of chemical space. In *Proceedings of the 34th International Conference on Machine Learning*, volume 70 of *Proceedings of Machine Learning Research*, pages 1470–1479, 2017.

[70] Benjamin J. Shields, Jason Stevens, Jun Li, Marvin Parasram, Farhan Damani, Jesus I. Martinez Alvarado, Jacob M. Janey, Ryan P. Adams, and Abigail G. Doyle. Bayesian reaction optimization as a tool for chemical synthesis. *Nature*, 590:89–96, 2021. doi: 10.1038/s41586-021-03213-y.

[71] V. Roshan Joseph. Space-filling designs for computer experiments: A review. *Qual. Eng.*, 28(1):28–35, 2016. doi: 10.1080/08982112.2015.1100447.

[72] A. Gilad Kusne, Heshan Yu, Changming Wu, Huairuo Zhang, Jason Hattrick-Simpers, Brian DeCost, Suchismita Sarker, Corey Oses, Cormac Toher, Stefano Curtarolo, Albert V. Davydov, Ritesh Agarwal, Leonid A. Bendersky, Mo Li, Apurva Mehta, and Ichiro Takeuchi. On-the-fly closed-loop materials discovery via Bayesian active learning. *Nat. Commun.*, 11:5966, 2020. doi: 10.1038/s41467-020-19597-w.

[73] David Eriksson, Michael Pearce, Jacob Gardner, Ryan D. Turner, and Matthias Poloczek. Scalable global optimization via local Bayesian optimization. In *Advances in Neural Information Processing Systems*, volume 32, 2019.

[74] Samuel Daulton, Maximilian Balandat, and Eytan Bakshy. Differentiable expected hypervolume improvement for parallel multi-objective Bayesian optimization. In *Advances in Neural Information Processing Systems*, volume 33, 2020.

[75] Ryan J. Tibshirani, Rina Foygel Barber, Emmanuel J. Candès, and Aaditya Ramdas. Conformal prediction under covariate shift. In *Advances in Neural Information Processing Systems 32*, pages 2530–2540, 2019.

[76] Clara Fannjiang, Stephen Bates, Anastasios N. Angelopoulos, Jennifer Listgarten, and Michael I. Jordan. Conformal prediction under feedback covariate shift for biomolecular design. *Proceedings of the National Academy of Sciences*, 119(43): e2204569119, 2022. doi: 10.1073/pnas.2204569119. URL `https://arxiv.org/abs/2202.03613`.